Extending Fisher–KPP glioma growth with oxygen and necrosis: a reproducible BrainWeb simulation and synthetic parameter recovery study

Working manuscript — 17 October 2026

Abstract

Background. Fisher–Kolmogorov–Petrovskii–Piskunov (Fisher–KPP) models describe glioma spread and growth but do not explicitly represent oxygen limitation or necrotic material. We developed a minimal coupled extension and an interactive workflow to examine these assumptions.

Methods. Tissue-dependent Fisher–KPP, oxygen-limited growth and viable-tumor–oxygen–necrosis dynamics were compared on BrainWeb over 90 days with assigned biological parameters. Analytical checks, ablations and refinement assessed the numerical method. Two synthetic experiments estimated homogeneous diffusion/proliferation and separate white-/gray-matter diffusion with shared proliferation from blurred, voxel-averaged signals. Early scans calibrated each fit; later scans were withheld. Each experiment used 100 bootstrap fits and 60 independent noise trials. BrainWeb inference additionally tested a change in anatomy/grid resolution.

Results. On the 2 mm forward grid, coupling reduced day-90 viable count by 14.80% versus Fisher–KPP, while low-threshold volume differed by 1.85%. Growth inhibition dominated; localized necrosis and oxygen minima remained grid sensitive. Homogeneous rates were recovered within bootstrap 95% intervals. In BrainWeb, known rates $(D_{\mathrm{WM}}, D_{\mathrm{GM}}, \rho) = (0.5, 0.1, 0.04)$ were estimated as $(0.49806, 0.08264, 0.03948)$ in $\mathrm{mm}^2/\mathrm{day}$, $\mathrm{mm}^2/\mathrm{day}$ and day^{-1}. The true proliferation rate fell just outside both 95% intervals in the primary realization. Independent local-interval coverage was 57/60, 55/60 and 59/60, respectively. Fitting noise-free 2 mm anatomy with the 3 mm model shifted gray-matter diffusion by +14.35%.

Conclusions. Oxygen feedback can alter the modeled interior despite similar threshold-defined extent. Conditional two- and three-rate inference is feasible under known initialization and imaging assumptions, but noise-based intervals omit anatomy/discretization error. Translation requires calibrated observation operators, broader uncertainty assessment, independent biological targets and external forecasts.

Keywords: glioma; reaction–diffusion; oxygen; necrosis; BrainWeb; parameter identifiability; MRI; validation.

1 Introduction

Reaction–diffusion models describe an expanding population through spatial dispersal and local growth. The Fisher–KPP equation emerged from Fisher and from Kolmogorov, Petrovskii and Piskunov [1, 2]. Related proliferation–invasion formulations describe glioma-cell migration through anatomically heterogeneous brain tissue. Distinguishing white- and gray-matter motility permits predictions of preferential spread even when local transport is isotropic [3].

A single viable-cell field omits microenvironmental processes that can distinguish an infiltrating margin from a poorly oxygenated interior. The histological illustration motivating this work,

from Comba et al., emphasizes invasion routes, vascular niches, reactive cells and necrosis [4]; the present model represents only a small subset of that conceptual picture. Earlier mechanistic modeling connected oxygen, cellular phenotypes and necrosis to pseudopalisading patterns [5]. Our reduction does not reproduce that model or explicitly resolve pseudopalisades. Oxygen-dependent proliferation and transfer into a necrotic compartment introduce additional assumptions and parameters. A more elaborate model can reproduce more patterns without providing uniquely identifiable biological explanations. Model complexity, observability and prediction therefore need to be assessed together.

Recent work illustrates the opportunities and constraints. Hiremath et al. investigated identifiability and selection among glioma treatment models using noisy virtual cohorts [6]. Miniere et al. incorporated serial MRI into a two-species model for 21 patients receiving chemoradiation, comparing sequential updates with prediction from limited imaging visits [7]. Zhang et al. explored physics-informed inference from a single structural MRI snapshot using characteristic nondimensional parameters [8]. These approaches motivate explicit consideration of what observations constrain; they do not establish that every mechanistic rate is identifiable from one scan.

Measurement validity is an additional issue. Independent histological evaluation of MRI-derived cellularity maps in 2025 found different performance in treatment-naive and recurrent gliomas [9]. A 2026 test–retest study combined diffusion, perfusion and oxygen-enhanced MRI to construct repeatable habitats in a murine glioma model [10]. Repeatability supports longitudinal inference, while biological specificity requires separate evidence. Recent 2026 preprints investigate poromechanical forecasting, sequential biomechanical model selection, and biopsy-derived kinetic information [11–13]. Their preprint status and experimental scope are retained here.

The present work develops a compact hierarchy from Fisher–KPP to oxygen-limited growth and coupled viable-tumor–oxygen–necrosis dynamics. Its contribution is a reproducible comparison on a fixed anatomical phantom, interactive exploration, numerical assessment, and staged synthetic recovery of homogeneous and tissue-specific rates with uncertainty, withheld-scan tests and anatomy sensitivity assessment. We ask how oxygen and necrosis alter the simulated interior relative to threshold-defined extent, which assumptions dominate that change, whether baseline and separate white-/gray-matter rates can be recovered under controlled observation assumptions, how discretized anatomy affects those estimates, and what measurements would be needed to interpret the parameters biologically. The study does not claim a new biological law or clinical forecasting performance.

2 Methods

2.1 Study design and anatomical data

The anatomical forward simulations used the original BrainWeb normal phantom, with its noise-free, intensity-uniform T1-weighted image and discrete tissue labels [14, 15]. The input is an anatomical substrate, not a longitudinal glioma dataset. No patient tumor masks, oxygen measurements or outcomes were available for calibration. The native 1 mm volumes contain $181 \times 217 \times 181$ voxels. The primary simulation used a 2 mm grid of $91 \times 109 \times 91$ voxels; intensities were resampled linearly and labels by nearest neighbor. Arrays use canonical RAS coordinates. Gray matter, white matter and the phantom's glial class form the transport domain Ω. CSF and other tissues are impermeable in every matched comparison. Data preparation and checksums are in Appendix A. Synthetic inference was evaluated first in a homogeneous phantom and then on the full BrainWeb anatomy. These experiments use a dilute known seed and synthetic observations; they do not calibrate the dense-seed coupled comparison.

2.2 Reference model: Fisher–KPP

Let $c(\boldsymbol{x}, t)$ denote viable tumor-cell density in cells/mm^3. The reference, M0, is

$$\partial_t c = \nabla\cdot(D(\boldsymbol{x})\nabla c) + \rho(\boldsymbol{x})c\left(1 - \frac{c}{K}\right), \qquad \boldsymbol{x} \in \Omega. \tag{1}$$

D is effective cell migration in mm^2/day, ρ is growth in day^{-1}, and K is a density scale. D is scalar with separate white- and gray-matter values; fiber-aligned transport is absent. Cell and, when present, oxygen transport have homogeneous no-flux boundaries. The matched reference uses the same seed, coefficients, anatomy and output times as the extension.

2.3 Oxygen-limited and three-field extensions

The coupled model, M2, adds relative oxygen availability $o(\boldsymbol{x}, t)$ and immobile necrotic material $z(\boldsymbol{x}, t)$ in tumor-cell equivalents/mm^3. Define

$$\phi = \frac{c+z}{K}, \quad g(o) = \min\left(1, \frac{o}{\theta_p}\right), \quad h(o) = \left[1 - \frac{o}{\theta_d}\right]_+, \quad p(\phi) = \frac{1}{1+\alpha\phi}, \tag{2}$$

where $[a]_+ = \max(a, 0)$ and $0 < \theta_d < \theta_p < 1$. The system is

$$\partial_t c = \nabla\cdot\left(D\nabla c\right) + \rho g(o)c\left[1-\phi\right]_+ - \delta h(o)c, \tag{3}$$

$$\partial_t z = \delta h(o)c - \lambda_z z, \tag{4}$$

$$0 = \ell_o^2\nabla^2 o + p(\phi)(1-o) - \beta(c/K)o. \tag{5}$$

Oxygen is solved at initialization and after every cell step. Its quasi-steady treatment imposes faster oxygen equilibration than tumor evolution, an assumption not tested by these data. Normal-tissue metabolism is implicit in the baseline $o = 1$. Distributed restoration replaces explicit vessels, and p is a phenomenological supply coefficient rather than measured blood flow. Viable cells consume oxygen. Necrotic material affects supply and available growth capacity but consumes no oxygen.

The two-field model M1 sets $\delta = 0$ and starts with $z = 0$, so oxygen only modulates growth. M2 includes death transfer and optional clearance. All examples start with $z = 0$. With $\beta = 0$, oxygen is one and M2 reduces to M0 for uniform K and the stated initial condition. Setting $\alpha = 0$ holds restoration fixed while retaining uptake. These nested cases provide implementation checks and ablations.

K regulates births rather than imposing hard packing on $c + z$. Diffusion can enter an already occupied location. Consequently, z/K and ϕ can exceed one: z/K is not a physical necrotic volume fraction. No healthy-cell population, edema, angiogenic state, tissue displacement, directed hypoxic migration or treatment response is included.

2.4 Initialization, parameters and numerical experiments

The initial density was a tissue-restricted Gaussian centered at RAS $(-30, 25, 30)$ mm, with standard deviations of 6 mm in each direction, a peak of 80,000 cells/mm^3, and truncation at three standard deviations. This dense seed was selected to produce an oxygen-limited interior early in the demonstration. Simulations cover 90 days, saving every 3 days. All biological values in Table 1 were assigned working values, not fitted estimates or established patient-level priors.

The primary M0–M2 comparison used $\Delta x = 2$ mm and maximum $\Delta t = 0.25$ day. Additional 2 mm runs evaluated M1, M2 with $\alpha = 0$, and M2 with $\beta = 0$. Temporal assessment repeated M2 with $\Delta t = 0.125$ and 0.0625 day. Native-grid assessment repeated M2 and matched M0 at

Table 1: Assigned parameters for the primary comparison. All are illustrative. Numerical tolerances and initialization details are in the appendices.

Symbol	Interpretation	Value	Unit
D_{WM}	White-matter migration	0.5	$\mathrm{mm}^2/\mathrm{day}$
D_{GM}	Gray/glial-matter migration	0.1	$\mathrm{mm}^2/\mathrm{day}$
ρ	Growth coefficient	0.04	day^{-1}
K	Density scale	100,000	$\mathrm{cells/mm}^3$
ℓ_o	Effective oxygen exchange length	0.2	mm
β	Relative tumor oxygen uptake	4	dimensionless
α	Supply suppression by total burden	4	dimensionless
θ_p	Oxygen for unrestricted proliferation	0.3	relative oxygen
θ_d	Hypoxic-death threshold	0.1	relative oxygen
δ	Maximum death-transfer rate	0.5	day^{-1}
λ_z	Necrotic-material clearance	0	day^{-1}

1 mm with a 0.125-day step. These nine model/step/grid cases are controlled simulations, not biological replicates. The grid comparison changes discretized anatomy as well as the numerical mesh, and is not a pure spatial-order experiment.

Cell transport uses conservative face fluxes with harmonic interface coefficients. Explicit diffusion is followed by exact frozen-coefficient logistic birth, hypoxic-death transfer and clearance substeps. Oxygen uses a compact no-flux stencil and preconditioned conjugate gradients. The method is first order in time. A diffusion positivity bound controls substeps but does not guarantee reaction accuracy. Details are in Appendix B.

2.5 Outcome measures and interactive visualization

We recorded viable count $N_c = \int_\Omega c\,\mathrm{d}V$, necrotic equivalents $N_z = \int_\Omega z\,\mathrm{d}V$, peak density, minimum tissue oxygen, and volumes above $c/K = 0.2$ and 0.8. Necrotic threshold volume uses $z/K \geq 0.2$; low-oxygen volume uses $o < \theta_d$. These are model-defined quantities, not validated MRI detection thresholds. With closed boundaries,

$$\frac{\mathrm{d}}{\mathrm{d}t}(N_c + N_z) = \int_\Omega \rho g(o) c\,[1-\phi]_+\ \mathrm{d}V - \lambda_z N_z. \tag{6}$$

Death transfer cancels. With zero clearance and matched initial states, the M0 viable-count minus M2 live-plus-dead-count difference measures the cumulative birth deficit within the numerical balance.

The Jupyter interface exposes biological and numerical controls and retains Fisher–KPP. Four fields share time and slice coordinates, with axial, coronal and sagittal views. Live 3D histories support arbitrary slicing; compact saved results retain seed-plane histories and final volumes. MP4/GIF exports share a frame schedule. A slower movie spans 20 seconds with a 6-second day-90 hold. Display settings do not alter fields. The paper uses lossless arrays and displays the full necrotic range.

2.6 Homogeneous recovery of diffusion and proliferation

To test two-parameter estimation independently of the anatomical comparisons, we solved normalized M0 dynamics, $u = c/K$, in a homogeneous 60 mm cubic no-flux domain. Ground-truth rates were $D = 0.300\ \mathrm{mm}^2/\mathrm{day}$ and $\rho = 0.045\ \mathrm{day}^{-1}$. The known initial Gaussian had peak 0.2, standard deviations $(5, 4, 5)$ mm and center $(-3, 2, 0)$ mm in box-centered coordinates. It was untruncated, with analytically integrated cell averages; this differs from the dense tissue-restricted BrainWeb seed. The synthetic signal applied a Gaussian point-spread function of 2 mm

standard deviation and averaged disjoint 6 mm cubic voxels. Independent zero-mean Gaussian noise, with known standard deviation 0.03 in normalized signal units, was added after imaging. Measurements were not clipped. This is a calibrated cellularity proxy, not a raw MRI signal or a claim that water ADC measures cell migration.

The generator used 1 mm cells and a maximum 0.0625-day step; fitting used 2 mm cells and a 0.25-day step with the same conservative diffusion/logistic kernel in float64. Different discretizations reduce numerical self-consistency bias, while the shared equations and observation model retain favorable structural assumptions. We measured their observation-space discrepancy explicitly. Scans at days 20, 40 and 60 supplied 3,000 scalar training measurements; days 75 and 90 were withheld from fitting and uncertainty estimation. The initial state, age origin, K, signal calibration, blur and noise were fixed and known.

Bounded Gaussian maximum likelihood estimated $(\log D, \log \rho)$ using nonlinear least squares [16], with three starting guesses and no probability prior. Local sensitivity-based marginal 95% confidence intervals were compared with percentile intervals from 100 parametric bootstrap fits [17]. Bootstrap parameter pairs generated pointwise confidence bands for the noise-free mean signal; adding independent measurement noise generated pointwise new-scan prediction intervals. A separate 60-dataset noise experiment, generated at the known rates on the fine grid, assessed recovery bias, RMSE and local-interval coverage with exact binomial uncertainty. This audits local intervals, not bootstrap coverage. A time-zero-only negative control tested rejection of unidentifiable rates. Equations, bounds, diagnostics and resampling details are in Appendix C.

2.7 Three-rate recovery on BrainWeb and anatomy sensitivity

The second inference experiment estimated D_{WM}, D_{GM} and a common ρ in normalized M0 dynamics on the full BrainWeb parenchyma. Ground truth was $(0.500, 0.100, 0.040)$, in $\mathrm{mm}^2/\mathrm{day}$, $\mathrm{mm}^2/\mathrm{day}$ and day^{-1}. Label 3 used D_{WM} and labels 2/8 used D_{GM}; CSF and extracerebral tissue were impermeable. No ordering or fixed ratio linked the diffusion rates. The known seed was an untruncated, point-sampled, tissue-restricted Gaussian of peak 0.2 and 6 mm widths at RAS $(-30, 25, 30)$ mm. The T1 image supplied anatomical context rather than measured tumor signals.

The primary generator and inverse model shared the full 3 mm tissue grid ($61 \times 73 \times 61$, 58,669 active voxels), with maximum steps 0.125 and 0.5 day, respectively. This isolates temporal mismatch but retains a shared spatial discretization; it is a favorable conditional recovery test. A fixed 60 mm observation cube centered at $(-30, 24, 30)$ mm supplied 10^3 voxels per scan. The PDE domain was not cropped. A known 2 mm Gaussian blur preceded physical-coordinate interpolation and approximate 6 mm voxel averaging. Independent Gaussian noise of known standard deviation 0.02 was added after imaging. All 3,000 measurements at days 20/40/60 were fitted; days 75/90 were withheld. Different noise levels and seeds prevent interpreting comparisons with the homogeneous experiment as an isolated effect of anatomy.

The three-log-rate likelihood, local covariance and percentile bootstrap followed the preceding procedure, with 100 bootstrap fits and 60 independent fine-time-step noise realizations. The latter audited local intervals, not bootstrap coverage. Three starts and sensitivity singular values checked local estimation. Forecasts were assessed over the full observation field of view (FOV) and a prespecified 18 mm radius around the known seed, containing 120 observation-voxel centers. This region was fixed before generating signal or noise.

For a separate sensitivity test, the generator used the full 2 mm BrainWeb grid with the same physical observation geometry, analytic seed and 0.125-day step. The 3 mm inverse model fitted both its noise-free signal and a noisy signal using the primary noise draw for a paired comparison. The test changes tissue geometry, mesh, seed sampling and discrete imaging; it does not isolate spatial order or estimate a distribution of segmentation errors. No confidence interval

was assigned to the deterministic noise-free rate shift. Appendix D supplies implementation and resampling details.

2.8 Proposed measurement-based estimation and validation protocol

Beyond the synthetic experiment, modality m at time t_j would be represented by

$$y_{mj} = \mathcal{H}_m[c_{\boldsymbol{\theta}}(t_j), o_{\boldsymbol{\theta}}(t_j), z_{\boldsymbol{\theta}}(t_j); \boldsymbol{\eta}_m] + \varepsilon_{mj}, \tag{7}$$

where $\mathcal{H}_m$ includes contrast, resolution, partial volume and the biological-state-to-measurement link. $\boldsymbol{\theta}$ includes biology and initialization; $\boldsymbol{\eta}_m$ contains observation parameters. Calibration must incorporate noise, registration and initial-state uncertainty. An objective and identifiability restrictions are in Appendix E.

The proposed sequence estimates a small growth–migration subset from serial MRI, introduces oxygen parameters when additional modalities support them, and calibrates cell-scale and necrosis mappings using independent samples. Numerical verification, synthetic recovery, withheld-time prediction, tissue agreement and external-cohort performance are evaluated separately. M0 and M2 must be fitted separately to identical calibration data before predictive comparison.

3 Results

3.1 Matched 90-day comparison

The coupled model produced a lower-density interior than Fisher–KPP, despite a similar spatial extent at the lower density threshold (Figure 1). At day 90, M2 contained 2.426 billion viable cells versus 2.847 billion in M0, a 14.80% reduction. The respective volumes with $c/K \geq 0.2$ were 48.832 and 49.752 mL, differing by only 1.85%. In contrast, the volume with $c/K \geq 0.8$ was zero in M2 and 1.944 mL in M0. Peak viable densities were 0.637 and 0.896 times K, respectively.

Minimum tissue oxygen in M2 was 0.0804. Necrotic material totaled 2.930 million cell equivalents, with $z/K \geq 0.2$ occupying 0.024 mL, or three 2 mm voxels. Low-oxygen volume ($o < 0.1$) was 0.104 mL. The dense initial seed was already oxygen limited; this experiment therefore does not demonstrate the spontaneous emergence of a macroscopic necrotic core from a dilute initial tumor. The tiny thresholded focus is not a resolved anatomical cavity. Moreover, the maximum $z/K = 1.226$ exceeds one, as permitted by the cell-equivalent formulation.

The viable-count gap between M0 and M2 was 421.24 million cells. Only 2.930 million cell equivalents were stored as necrotic material. With no clearance, the remaining 418.31 million difference in live-plus-dead burden is attributable to reduced cumulative births in the discrete balance. Thus, in this case, the smaller viable burden mainly reflects inhibited proliferation rather than death transfer.

3.2 Controlled model ablations

Table 2 and Figure 2 separate growth inhibition, death transfer and the chosen supply feedback. Removing hypoxic death (M1) increased day-90 viable count by approximately 0.14% relative to M2. In contrast, removing burden-dependent suppression of oxygen supply ($\alpha = 0$) gave 2.829 billion viable cells, approximately 0.62% below M0. Its minimum oxygen, 0.2245, remained above the death threshold, and no necrotic material formed. The supply closure therefore has a much larger effect on this particular comparison than the death term. This finding is conditional on the assigned parameters and dense seed, and does not establish the closure biologically.

Setting uptake to zero ($\beta = 0$) maintained $o = 1$, yielded no necrotic material and reproduced the M0 solution. The largest final density discrepancy divided by K was below 10^{-5}, consistent

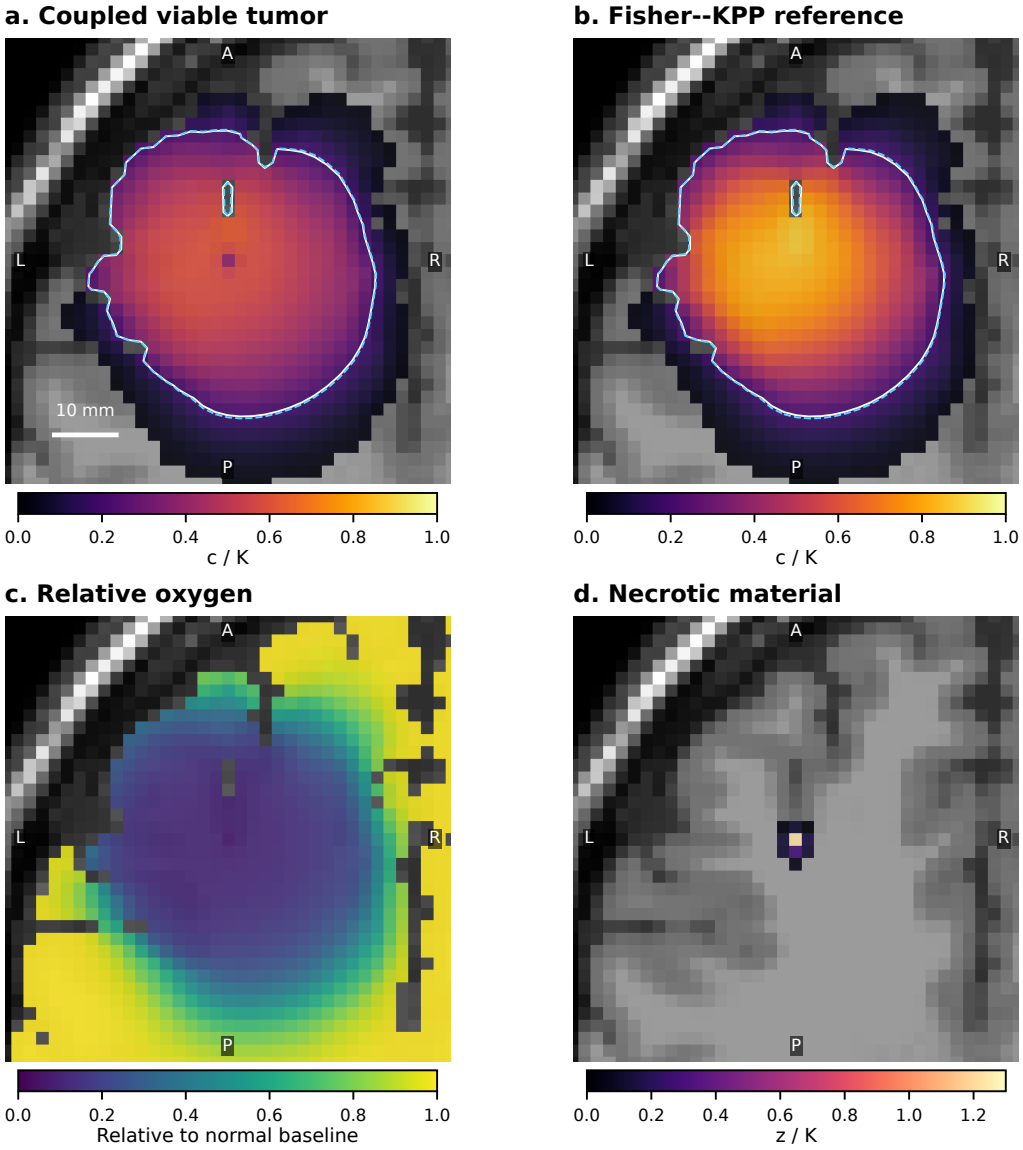


Figure 1: Day-90 fields in the matched primary simulations. Axial views at RAS $z = 30$ mm show a 72×72 mm region, with the original 2 mm simulation pixels retained. Viable density and the reference share the same 0–1 color scale; oxygen is relative to the normal baseline. Necrotic material is shown over 0–1.3 without clipping its peak at one. White solid and cyan dashed contours mark $c/K = 0.2$ in M2 and M0, respectively. Tumor/necrosis overlays below 0.02 of K are transparent for display only. The scale bar is 10 mm. These thresholds and colors have no calibrated correspondence to MRI contrast.

Table 2: Day-90 outcomes for 2 mm model ablations with a maximum 0.25-day step. $V_{0.2}$ and $V_{0.8}$ are viable-density threshold volumes in mL. Oxygen and necrotic state variables are absent from M0, indicated by dashes.

Model	$N_c/10^9$	$V_{0.2}$	$V_{0.8}$	$\min o$	$N_z/10^6$
M0: Fisher–KPP	2.847	49.752	1.944	—	—
M1: oxygen-limited growth	2.429	48.856	0.000	0.0962	0.000
M2: oxygen and necrosis	2.426	48.832	0.000	0.0804	2.930
M2, fixed supply ($\alpha = 0$)	2.829	49.744	0.976	0.2245	0.000
M2, zero uptake ($\beta = 0$)	2.847	49.752	1.944	1.0000	0.000

Table 3: Day-90 outcomes under temporal and anatomical-grid refinement for M2. h is voxel spacing in mm and Δt is the maximum step in days. The 1 mm run changes both the numerical mesh and the discretized tissue boundaries.

h	Δt	$N_c/10^9$	$N_z/10^6$	$\min o$	$\max(z/K)$
2	0.25	2.4258	2.9302	0.0804	1.226
2	0.125	2.4239	2.9431	0.0798	1.266
2	0.0625	2.4230	2.9493	0.0795	1.285
1	0.125	2.4524	3.6067	0.0513	3.416

Table 4: Recovery from the first three noisy scans of the homogeneous phantom. Local intervals use the known-noise residual Jacobian; bootstrap intervals are percentiles of 100 refits. Intervals are marginal, frequentist and conditional on known initialization and imaging assumptions.

Parameter	Known	Estimate	Local 95% CI	Bootstrap 95% CI
D (mm^2/day)	0.30000	0.30326	[0.27217, 0.33791]	[0.27257, 0.33507]
ρ (day^{-1})	0.04500	0.04510	[0.04413, 0.04609]	[0.04430, 0.04604]

with finite-precision differences between the numerical paths. This is a limiting-case verification, not an independent biological comparison.

3.3 Numerical verification and resolution dependence

The software test suite passed all 50 tests. Checks include analytical logistic behavior, diffusion conservation, boundary behavior, positivity, death-transfer balance, oxygen bounds and model limits (Appendix B). These results verify implemented properties without validating the biological assumptions.

At 2 mm, reducing the maximum time step from 0.25 to 0.125 day decreased the final relative L^2 difference in viable density against the 0.0625-day calculation from 0.1270% to 0.0423%. Corresponding necrotic-field differences decreased from 4.238% to 1.386%. The relative oxygen differences were 0.0129% and 0.0043%. These comparisons support temporal refinement of the implemented scheme; a finite reference calculation does not provide an exact error or establish an asymptotic convergence order.

The native 1 mm calculation contained 2.452 billion viable cells, compared with 2.878 billion in its matched M0 reference, a 14.78% reduction. This relative reduction is similar to the primary result. However, minimum oxygen fell to 0.0513, the necrotic-equivalent count increased to 3.607 million and peak z/K rose to 3.416. The necrotic threshold occupied 0.032 mL (32 native voxels), while low-oxygen volume was 0.027 mL. These local quantities are sensitive to spatial resolution and to the discrete anatomy. Neither grid resolves the assigned 0.2 mm oxygen length; spatial convergence of the oxygen–necrosis subsystem has not been demonstrated.

3.4 Homogeneous parameter recovery and uncertainty

In the separate homogeneous experiment, all three initial guesses converged to the same fit at the reported precision. Estimated diffusion was 0.3033 mm^2/day and proliferation was 0.04510 day^{-1}, relative errors of +1.09% and +0.22%, respectively (Table 4). Both known values lay within the local and bootstrap marginal 95% intervals. The log-parameter correlation was +0.540, indicating that uncertainty in the two rates should be propagated jointly. No primary or replicate fit reached a bound or failed. Agreement across starts and finite sensitivity singular values support local estimation for this experiment; they do not establish global identifiability under unknown initialization.

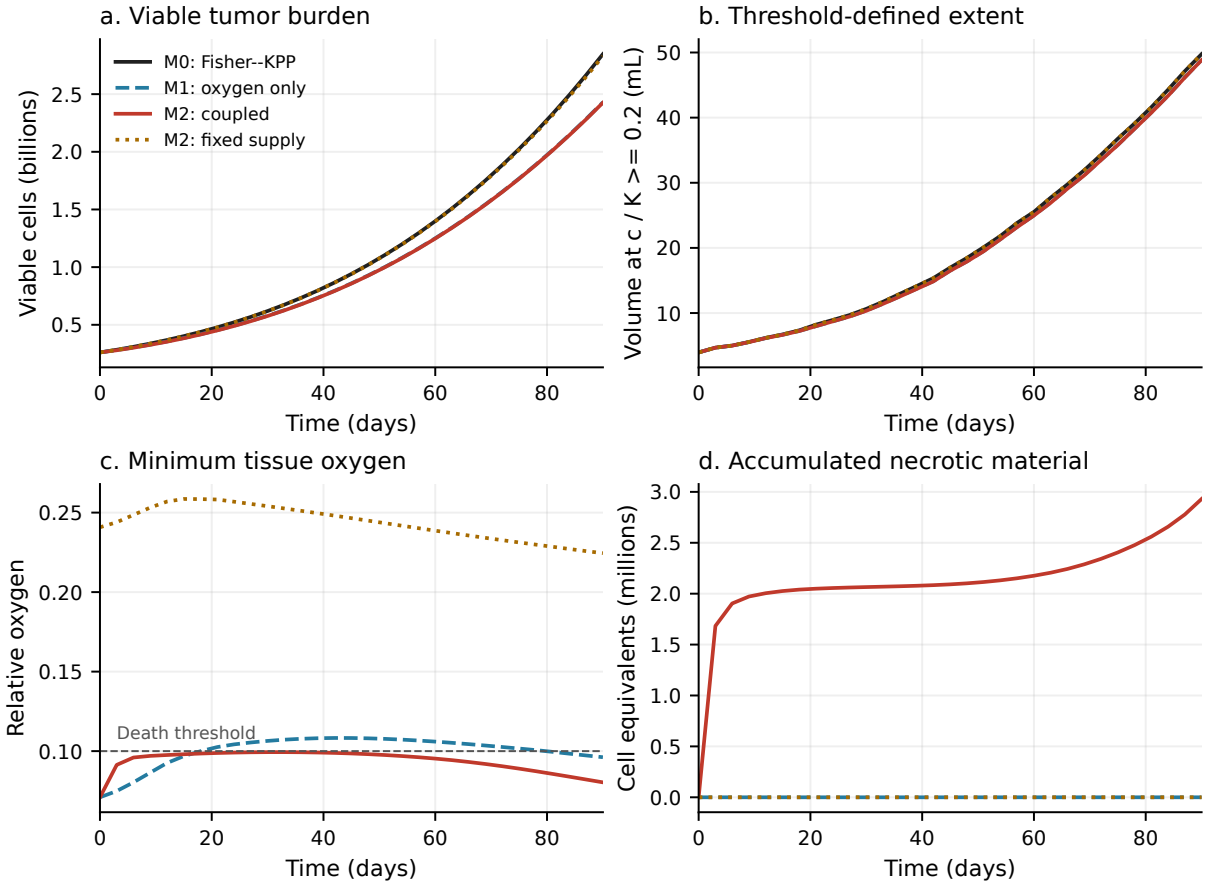


Figure 2: Time courses for the matched reference and selected ablations. Panels show viable cell count, threshold-defined viable volume, minimum tissue oxygen and necrotic cell equivalents. M0 has no oxygen or necrosis state; these curves are consequently omitted. The dashed horizontal line in panel c marks $\theta_d = 0.1$. M1 and M2 burden curves nearly overlap. All curves are deterministic simulations with assigned parameters; they are not measurements, replicate averages or uncertainty intervals.

Table 5: Independent repeated-noise recovery audit of local 95% intervals. Bias and RMSE retain the units in the parameter column. The last column quantifies sampling uncertainty in empirical coverage, not uncertainty in an individual parameter estimate.

Parameter	Bias	RMSE	Coverage	Binomial 95% CI
D (mm^2/day)	$+2.64 \times 10^{-3}$	1.71×10^{-2}	59/60 (98.3%)	[91.1, 100.0]%
ρ (day^{-1})	$+3.60 \times 10^{-7}$	4.48×10^{-4}	60/60 (100.0%)	[94.0, 100.0]%

Table 6: Withheld synthetic-scan forecast errors in normalized signal units, evaluated over all 1,000 observation voxels per scan. Prediction intervals are approximate and pointwise, conditional on the fitted model.

Day	RMSE vs clean truth	RMSE vs noisy scan	Voxel fraction in 95% prediction interval
75	4.72×10^{-4}	0.03036	94.0%
90	8.42×10^{-4}	0.03018	94.6%

Across 60 independent noise realizations, local intervals covered D in 59/60 trials and ρ in 60/60 trials (Table 5; Figure 3). The exact binomial intervals include the nominal 95% rate and remain broad. The observed 98.3% and 100% fractions therefore neither establish exact calibration nor imply guaranteed coverage. The study varies measurement noise at one fixed biology, initialization and imaging setting, rather than sampling a virtual patient population. The time-zero-only control was correctly rejected as locally unidentifiable.

3.5 Withheld homogeneous forecasts and numerical discrepancy

Day-75 and day-90 predictions used no observations beyond day 60. Whole-image RMSE against the known noise-free signal was 4.72×10^{-4} and 8.42×10^{-4}, respectively, while RMSE against the withheld noisy images was approximately the 0.03 noise standard deviation (Table 6). The 95% new-scan prediction intervals contained 94.0% and 94.6% of the withheld voxel measurements. These are spatial fractions in two scans, not coverage estimates from independent patients; the whole-box metrics include background voxels. Figure 3b separately displays the signal near the tumor. Confidence in the mean signal is narrower than prediction uncertainty for a future noisy measurement.

At day 90, the observation-space RMSE between the inference grid and fine generator at the known parameters was 3.41×10^{-4}, approximately 1.14% of the noise standard deviation. Halving the inference time step changed the signal by RMSE 6.19×10^{-5}. The finer generator is still a numerical reference, and a small per-voxel discrepancy does not by itself ensure negligible bias when many voxels are used; the repeated-noise audit supplies a complementary check.

3.6 BrainWeb tissue-specific parameter recovery

The three starting guesses agreed to the reported precision. The estimated $(D_{\mathrm{WM}}, D_{\mathrm{GM}}, \rho)$ was $(0.49806, 0.08264, 0.03948)$, with relative errors -0.39%, -17.36% and -1.30%, respectively (Table 7). Both diffusion truths lay within the local and bootstrap 95% intervals. The true $\rho = 0.040$ day^{-1} fell just above their respective upper limits, 0.039968 and 0.039920 day^{-1}. This noise realization and its interval misses were retained. No primary, bootstrap or independent-trial fit failed or reached a bound.

Gray-matter diffusion had the widest relative uncertainty. The log-rate correlation between white- and gray-matter diffusion was -0.582 (Figure 4a). Weighted sensitivity singular values were approximately 184.35, 24.64 and 5.25, with Jacobian condition number 35.14. These

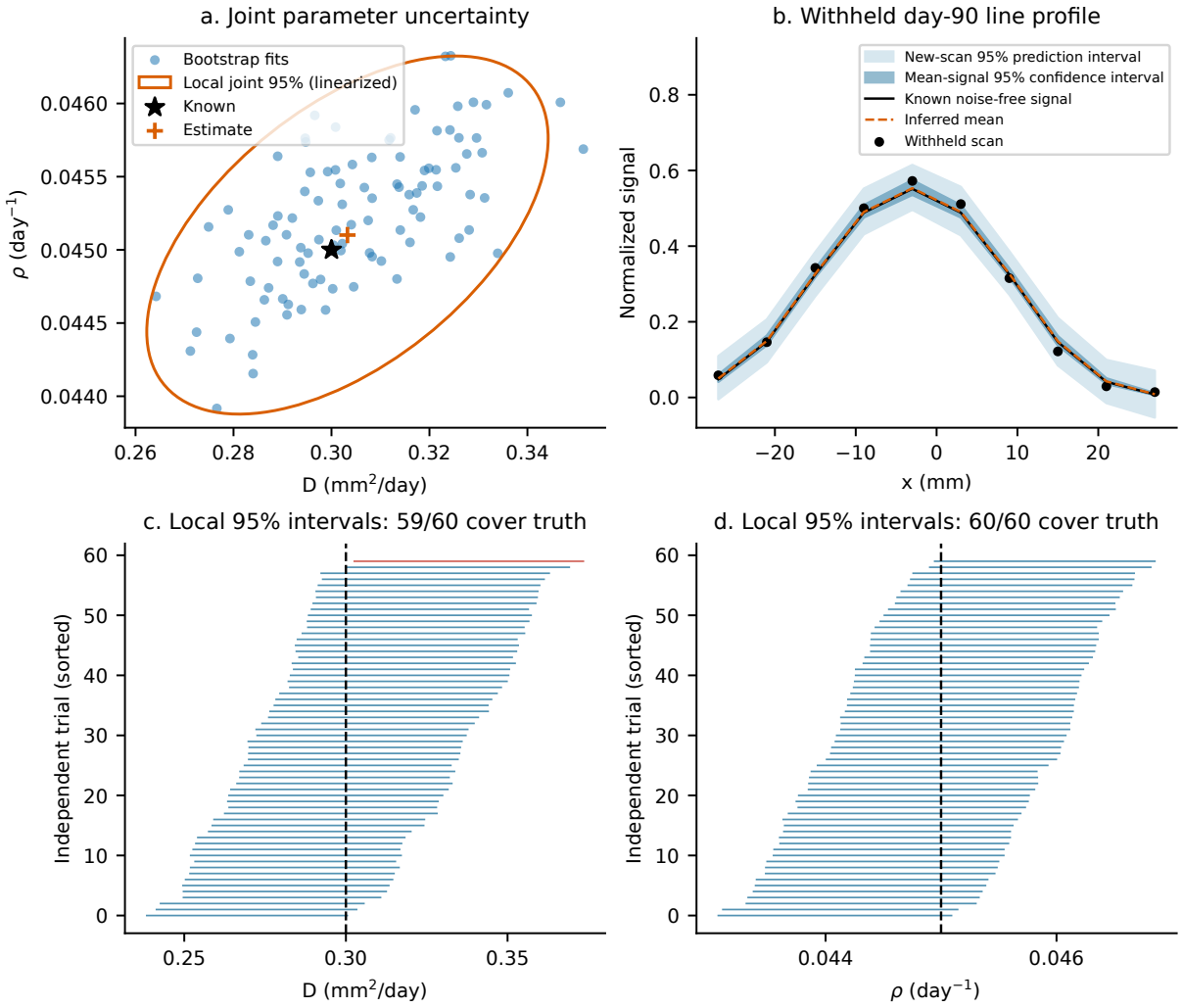


Figure 3: Synthetic recovery and conditional uncertainty. (a) Joint bootstrap estimates and a local joint 95% ellipse, linearized into physical units. The star marks the known rates. (b) A day-90 line profile through the nearest observation-voxel centers to the seed's y, z coordinates. The darker mean-signal confidence band accounts for parameter uncertainty; the lighter new-scan prediction band also includes independent noise. Both are pointwise intervals. (c,d) Local marginal 95% intervals for the 60 independent noise realizations, sorted by fitted rate. Red denotes a missed known value; the dashed line is ground truth. These trials audit local intervals, not bootstrap coverage or clinical variability.

Table 7: Three-rate recovery from noisy BrainWeb signals. Units are mm^2/day for diffusion and day^{-1} for proliferation. Intervals are marginal and conditional on the known seed, anatomy, imaging model and noise. The true proliferation rate is outside both intervals in this realization.

Parameter	Known	Estimate	Local 95% CI	Bootstrap 95% CI
D_{WM} (mm^2/day)	0.50000	0.49806	[0.45121, 0.54978]	[0.45033, 0.54892]
D_{GM} (mm^2/day)	0.10000	0.08264	[0.05714, 0.11952]	[0.05543, 0.10749]
ρ (day^{-1})	0.04000	0.03948	[0.03899, 0.03997]	[0.03886, 0.03992]

Table 8: BrainWeb repeated-noise audit of local 95% intervals. Bias and RMSE retain the units shown. Binomial intervals describe Monte Carlo uncertainty in the coverage fraction, not uncertainty in a fitted rate.

Parameter	Bias	RMSE	Coverage	Binomial 95% CI
D_{WM} (mm^2/day)	$+2.21 \times 10^{-3}$	2.44×10^{-2}	57/60 (95.0%)	[86.1, 99.0]%
D_{GM} (mm^2/day)	$+1.10 \times 10^{-3}$	1.86×10^{-2}	55/60 (91.7%)	[81.6, 97.2]%
ρ (day^{-1})	-1.13×10^{-5}	2.06×10^{-4}	59/60 (98.3%)	[91.1, 100.0]%

Table 9: Withheld BrainWeb forecast metrics. The full FOV contains 1,000 observation voxels; the near-seed region contains 120. Errors use normalized signal units. The final column is the spatial fraction of noisy observations in approximate pointwise 95% new-scan prediction intervals.

Day	Region	RMSE vs clean	RMSE vs noisy	Voxel fraction in prediction interval
75	Full FOV	0.00227	0.02004	94.9%
75	Near seed	0.00496	0.02056	95.0%
90	Full FOV	0.00372	0.02064	94.2%
90	Near seed	0.00730	0.02344	89.2%

local diagnostics support estimating three rates for this specific design, without proving global identifiability or robustness to uncertain initialization. Time-zero-only data were correctly rejected. A separate unit test rejected two-tissue inference in an all-white-matter domain.

Across 60 independent noise datasets, local 95% interval coverage was 57/60 for D_{WM}, 55/60 for D_{GM} and 59/60 for ρ (Table 8; Figure 4d–f). Exact binomial intervals include 95% but are broad. Repeated-noise biases were small relative to single-fit uncertainty; the audit neither guarantees nominal coverage nor evaluates bootstrap coverage. The primary proliferation miss is compatible with sampling variation rather than grounds for replacing the noise seed.

3.7 BrainWeb withheld forecasts and anatomy sensitivity

Day-90 RMSE against the noise-free signal was 3.72×10^{-3} over the full FOV and 7.30×10^{-3} in the prespecified near-seed region (Table 9). New-scan prediction intervals contained 94.2% of all withheld day-90 voxels but only 89.2% of near-seed voxels. Thus, background-inclusive summaries can conceal less favorable performance near the tumor. These are descriptive spatial fractions for one noisy scan, not independent coverage trials or simultaneous spatial guarantees.

On the shared 3 mm anatomy, changing the maximum step from 0.5 to 0.25 day reduced day-90 signal RMSE relative to the 0.125-day generator from 1.50×10^{-4} to 5.01×10^{-5}. By contrast, the 3 mm versus 2 mm generator discrepancy at the known rates reached 3.68×10^{-3} at day 90, or 18.4% of the 0.02 noise standard deviation (Figure 5). This is a change in discretized anatomy and numerical representation rather than a pure spatial-convergence comparison.

Fitting the noise-free 2 mm signal with the 3 mm model estimated $D_{\mathrm{GM}} = 0.114350$ mm^2/day, a +14.35% shift from the known value (Table 10). Corresponding white-matter and proliferation shifts were +0.04% and -0.13%. The paired noisy stress fit gave $(0.499160, 0.096770, 0.039445)$ in the respective units. Its conditional local intervals do not incorporate anatomy/grid discrepancy. Signal error below one noise standard deviation per voxel can therefore coexist with a material rate shift when systematic differences are fitted across many measurements. This sensitivity example does not estimate segmentation uncertainty or establish a universal gray-matter bias.

Both recovery experiments fit synthetic M0 observations only. The oxygen–necrosis parameters and the dense-seed forward comparisons remain assigned inputs; no patient parameters, oxygen/death rates or clinical predictive errors were estimated.

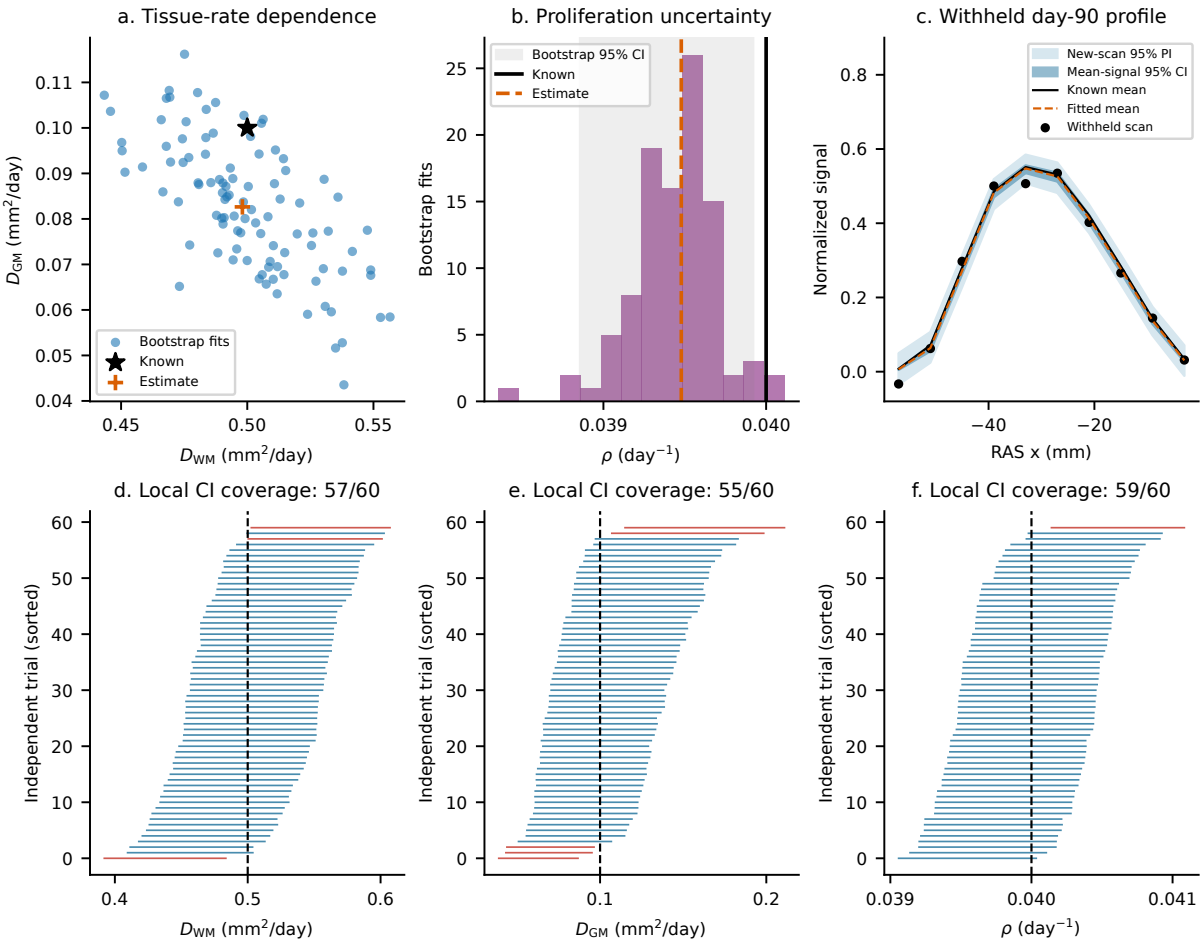


Figure 4: BrainWeb recovery and conditional uncertainty. (a) Joint white-/gray-matter bootstrap estimates; the star denotes known rates and the cross the primary estimate. (b) Proliferation bootstrap distribution, with marginal percentile interval shading; its known value is just beyond the upper limit. (c) Withheld day-90 profile through the nearest observation-voxel centers to the seed's y, z coordinates. Mean-signal confidence and new-scan prediction intervals are pointwise and have different meanings. (d–f) Local 95% intervals for 60 independent noise realizations, sorted by fitted rate. Red marks a missed known value. These repetitions share anatomy and biology and do not constitute a virtual patient cohort.

Table 10: Anatomy/grid sensitivity: full 2 mm data fitted with the 3 mm model. The noiseless fit isolates a deterministic rate shift under the changed representation. The noisy fit reuses the primary observation-noise draw. Neither fit estimates a distribution of anatomy uncertainty.

Parameter	Known	Noiseless fit	Relative shift	Paired noisy fit
D_{WM} (mm^2/day)	0.50000	0.500218	+0.04%	0.499160
D_{GM} (mm^2/day)	0.10000	0.114350	+14.35%	0.096770
ρ (day^{-1})	0.04000	0.039948	-0.13%	0.039445

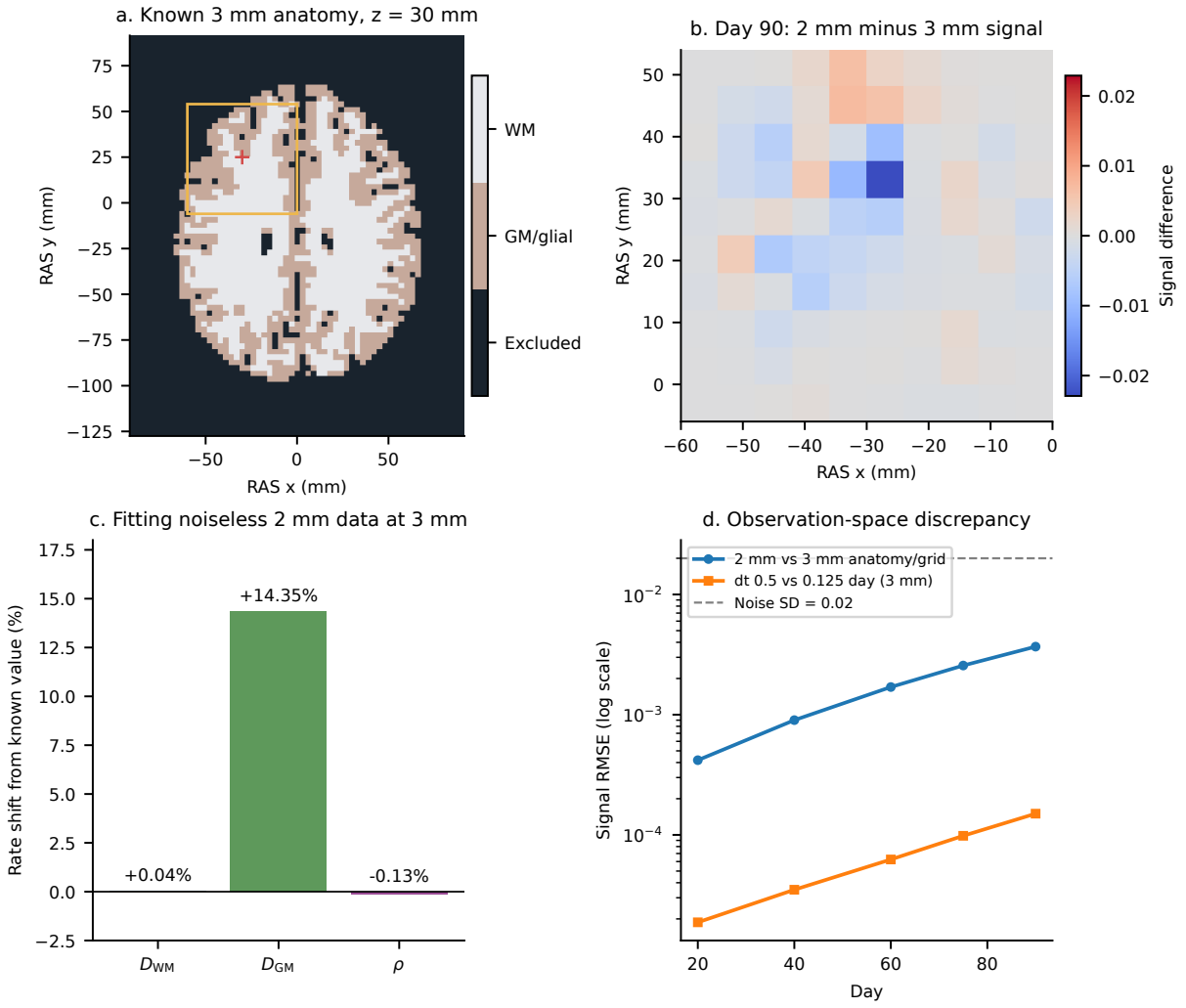


Figure 5: Sensitivity to anatomy and numerical representation. (a) Full 3 mm tissue domain at RAS $z = 30$ mm, with the known seed center and fixed 60 mm observation FOV. The forward PDE is not cropped to that box. (b) Difference between 2 mm and 3 mm noise-free generating signals at day 90, using identical physical observation voxels; the displayed observation plane is at $z = 27$ mm. (c) Relative shifts after fitting the noise-free 2 mm data with the 3 mm inverse model. (d) Observation-space discrepancies at the known rates; temporal error on the shared 3 mm grid is smaller than the combined anatomy/grid effect. Both generators in the anatomy comparison use a 0.125-day step. No biological measurement or clinical validation is represented.

4 Discussion

4.1 Interpretation and scope of the extension

The principal result is a separation between simulated burden and threshold-defined extent. With identical assigned migration and growth parameters, oxygen feedback lowered viable density in the interior while leaving the lower-density boundary comparatively similar. Well-oxygenated regions retain Fisher–KPP-like growth, whereas the interior experiences inhibited births and, at sufficiently low oxygen, death transfer. The ablations show that burden-dependent supply suppression dominates the overall effect for this parameter set. The numerical comparison establishes how the selected equations behave; it does not establish that they predict glioma better than Fisher–KPP. Predictive model selection requires fitting each candidate to the same calibration observations and comparing their withheld-data predictions.

The extension is deliberately small, but its state interpretation matters. Immobile z with zero clearance accumulates historical cell loss, while viable cells can diffuse into the same locations. The equations therefore allow $c + z > K$ and $z > K$ locally. A normalized necrotic volume fraction should not be inferred from z/K. The observed grid sensitivity reinforces this limitation. If physical packing, tissue displacement or a necrotic cavity becomes a target, a volume-fraction, mechanical or poromechanical formulation would be a different model requiring additional parameters and observations. Recent MRI-informed poromechanical forecasting and sequential biomechanical model-selection studies provide relevant alternatives, but the cited 2026 records are preprints and concern distinct preclinical datasets [11, 12]. They do not validate the present closure.

4.2 What the synthetic recovery establishes

The two recovery experiments provide a staged inference test: homogeneous diffusion/proliferation and then separate white-/gray-matter diffusion with shared proliferation were estimated from noisy, low-resolution signals. Both include local and bootstrap intervals, independent noise trials and withheld future scans. The homogeneous generator used a finer mesh and time step than inversion. The primary BrainWeb generator instead shared the spatial grid while refining time; its separate anatomy/grid stress test exposes a limitation of that favorable design. Neither experiment removes shared biological, initialization or imaging assumptions, and neither estimates coupled oxygen–necrosis mechanisms from structural MRI.

The homogeneous known rates lay within both interval constructions. In the BrainWeb primary realization, the true proliferation rate narrowly missed both 95% intervals, while both diffusion rates were contained. Retaining that outcome is essential: a nominal confidence level does not imply containment in every realized dataset. Independent local-interval coverage of 57/60, 55/60 and 59/60 had broad binomial uncertainty. These observations do not establish bootstrap coverage, global identifiability or calibration across patients. The weaker relative precision of gray-matter diffusion and its dependence on white-matter diffusion argue for propagating the joint rate uncertainty. Separate tissue rates were estimable here without fixing their ratio, but that conclusion depends on this known seed, informative tissue exposure and observation schedule. Relaxing initialization and calibration may expose stronger confounding [6].

The anatomy test identifies a distinct source of uncertainty. Noise-free 2 mm signals fitted with the 3 mm model shifted gray-matter diffusion by +14.35%, despite day-90 observation discrepancy remaining below the noise standard deviation. Many measurements can make sampling intervals narrow while systematic representation error persists. This shift combines tissue-boundary resampling, mesh, seed sampling and discrete imaging; it is not a measured biological change, a pure spatial-order result or an estimated segmentation-error distribution. The primary bootstrap and local covariance keep these quantities fixed and cannot quantify

this effect. Inference should therefore compare multiple anatomically consistent resolutions and propagate plausible label/registration variations before interpreting tissue-rate differences biologically.

Forecast assessment also needs spatial qualification. In BrainWeb, day-90 prediction intervals contained 94.2% of full-FOV measurements but 89.2% near the seed. Background-inclusive summaries alone can conceal local mismatch. Mean-signal confidence bands and new-measurement prediction bands answer different questions; neither provides simultaneous image coverage here. These frequentist conditional intervals are not Bayesian posteriors or estimates of clinical variability. Next studies should vary seed location, tissue-rate contrast and scan timing, and introduce unknown age, seed shape, capacity, calibration, correlated noise and structural mismatch. A broader bootstrap-coverage audit remains separate work.

4.3 What measurements could identify parameters?

A practical first stage would estimate a limited growth–migration subset from serial, registered MRI, while jointly assessing initialization and whether separate tissue migration rates are supported. A constrained tissue ratio or an external prior may be needed when the measurements do not separate those rates; the conditional BrainWeb example does not establish that this restriction can be removed in patients. In a homogeneous, isotropic Fisher–KPP medium, the asymptotic minimal front speed is $v = 2\sqrt{D\rho}$ and the front length scale is proportional to $L = \sqrt{D/\rho}$. These relations explain why growth velocity and spatial spread offer complementary information. They are not identities for a finite, heterogeneous brain with an unknown initial state. One snapshot with unknown tumor age generally cannot separate all dimensional rates. Recent single-scan inference methods introduce model and scaling assumptions [8]; a 2026 simulated-biopsy preprint similarly distinguishes information about infiltration length from information about growth speed [13].

MRI-derived cell-density maps can supply more information than binary segmentations only when their calibration is defensible. ADC measures water mobility, not the effective migration coefficient D. Its association with cellularity is influenced by extracellular space, edema, tissue composition and necrosis [18]. Nor should a FLAIR boundary automatically be treated as a fixed tumor-density contour: an MRI–histology study challenged that assumption in a single autopsy case [19]. In a 2025 study of image-guided biopsies, associations between MRI-derived cell-density maps and histology differed between untreated and recurrent tumors [9]. The treatment context and observation mapping must therefore be part of the inference model. Biopsy coordinates, brain shift, fixation deformation, partial volume and the conversion from two-dimensional counts to cells/mm^3 require explicit treatment.

Oxygen introduces further observation ambiguity. The state o is availability relative to a normal baseline; it is not directly oxygen extraction fraction, hemoglobin saturation or oxygen tension. Oxygen-enhanced MRI, BOLD-based measures and hypoxia PET could constrain an observation model, but their signals depend on delivery, vascular and tissue factors. A histological comparison of MRI-derived oxygen measures illustrates that an imaging surrogate cannot simply be equated with a hypoxia marker [20]. The 2026 glioma habitat repeatability study is encouraging for measurement consistency, but repeatability does not establish an absolute oxygen calibration or biological specificity [10].

Priors are necessary where these measurements leave parameters weakly identified. Positive rates and lengths can use log-normal priors; ordered oxygen thresholds can use a transformed prior satisfying $0 < \theta_d < \theta_p < 1$. Hyperparameters should come from independently evaluated experiments or a defensible population model, with uncertainty reflecting transfer between systems. The working values in Table 1 are not empirical prior means. Prior predictive checks should reveal implausible growth, oxygen depletion and accumulation above the intended interpretation of K.

Table 11: Proposed measurement and prior strategy. These are candidate information sources, not measurements acquired or estimates obtained in this study. Each modality requires its own observation operator and uncertainty model.

Quantity	Potential information	Restriction or prior needed
$c(0)$, D, ρ	Serial T1/T2/FLAIR, diffusion-weighted and quantitative MRI; registered anatomy	Joint initialization uncertainty; estimate a small parameter subset; constrain tissue ratios. MRI water diffusion is not cell migration.
K and cell scale	Spatially localized tumor-cell counts with tissue-volume or stereological calibration	Distinguish tumor from non-tumor cells and correct sampling/deformation; MRI alone does not fix the absolute scale.
β, α	Oxygen-sensitive imaging combined with perfusion/vascular measurements and externally calibrated uptake information	Supply and consumption are confounded; blood-flow surrogates are not the restoration coefficient itself.
ℓ_o	External transport measurements or multiscale vascular/oxygen models	Constrain externally when subvoxel; the current 1–2 mm meshes do not resolve 0.2 mm.
θ_p, θ_d, δ	Controlled oxygen-dependent growth and survival experiments; spatially matched hypoxia/viability observations	Relative-oxygen scale must be calibrated. Ki-67 is a cycling fraction, not directly a growth rate; hypoxia markers are not absolute oxygen readings.
λ_z and z mapping	Longitudinal necrotic-region imaging plus localized pathology	Separate formation from clearance and cell equivalents from tissue volume; a single scan cannot determine both histories.

Sensitivity analysis, profile likelihoods and posterior correlations can then identify parameter combinations that the observations fail to distinguish. The 2025 identifiability/model-selection study provides a relevant framework, although its virtual-cohort chemoradiation models differ from this system [6]. A narrow posterior produced mainly by a strong prior should not be described as precise measurement-based estimation.

4.4 Uncertainty quantification and informative measurements

The positive rate correlation and forecast ensembles in the synthetic experiment illustrate why parameter estimation should quantify joint uncertainty rather than supply a single best fit. Beyond the conditional frequentist analysis performed here, a Bayesian analysis could combine independently justified priors with an observation likelihood accounting for imaging noise and spatial correlation. Hierarchical population priors could support patients with sparse imaging, provided the population model is learned within training data. Uncertain initialization, registration and segmentation should be included as nuisance quantities or evaluated through repeated analyses. Strong posterior correlations and sensitivity to alternative priors would indicate which rates or combinations remain poorly constrained.

Forecast ensembles should propagate these uncertainties into viable-burden intervals, infiltration maps and oxygen/necrosis predictions. Measurement uncertainty, numerical error and model inadequacy require separate assessment: a narrow conditional posterior does not account automatically for a wrong oxygen closure or unresolved spatial gradients. Competing Fisher–KPP and coupled formulations should be evaluated separately on withheld observations; any model-averaging weights should be learned without those evaluation data. Posterior predictive checks, interval coverage and width, and sensitivity to the observation operator should accompany point-error metrics.

Pash et al. developed a scalable Bayesian reaction–diffusion framework using longitudinal imaging and investigated imaging frequency and experimental design [21]. This provides a methodological reference for the proposed inference workflow, not validation of the present oxygen–necrosis extension. Once uncertainty is calibrated, candidate scan times or additional modalities could be ranked by their expected reduction in forecast uncertainty or their expected value for distinguishing treatment options. Additional measurements are useful when they resolve a consequential ambiguity, rather than merely increasing the number of fitted parameters.

4.5 A validation pathway with independent targets

Validation must be tied to a specified observable, population and prediction horizon. Five complementary stages are appropriate here.

1. **Verify the solver.** Check conservation, positivity, analytical limits, solver tolerances and time/space refinement. These checks address mathematical implementation. The present unresolved oxygen length and necrotic peak require further spatial assessment before local quantitative claims.
2. **Extend synthetic recovery.** The homogeneous and BrainWeb experiments supply controlled two- and three-rate recovery and withheld-scan checks. Broaden them across seed locations, rates, anatomy representations and scan schedules, and add unknown initialization, correlated noise, registration error and observation-model mismatch. Use a different mesh, solver or observation formulation to reduce the inverse crime of fitting data generated by the identical numerical pipeline. Synthetic success does not establish biological adequacy; earlier MRI-constrained reaction–diffusion work demonstrated the importance of separating synthetic recovery from in vivo forecasting [22].
3. **Forecast withheld observations.** Fit early observations, freeze the calibration and measurement mapping, and predict later scans over a declared horizon. Compare separately calibrated

M0, M1 and M2 with a simple growth baseline. Treated tumors require explicit treatment terms or a justified untreated interval. Updating with each new scan is sequential assimilation, a different evaluation from a frozen forecast. The 2025 assimilation study illustrates the distinction in high-grade glioma during chemoradiation [7].

4. **Check latent fields independently.** Reserve spatially localized cell counts, viable/necrotic histology and oxygen-related observations for evaluation rather than using all samples for calibration. Report localization and measurement uncertainty. A successful volume forecast alone does not validate o, z or the death mechanism. Controlled preclinical oxygen or perfusion perturbations would provide stronger tests of the proposed feedback than morphology alone.

5. **Evaluate transportability.** Apply a fixed analysis and prior strategy to an external cohort, with any permitted patient-specific calibration specified in advance. Group splits by patient and, where relevant, site and treatment status. Prospective clinical utility is a subsequent question, beyond retrospective image agreement.

Evaluation should include volume errors, surface distances and overlap, together with quantitative field errors where calibrated measurements exist. Report predictive interval coverage and width, not only point accuracy. Repeated biopsies within a patient and spatially correlated voxels are not independent replicates; patient-level or hierarchical analyses should reflect this dependence. The 2025 biopsy-validation study is a useful example of independent tissue evaluation and context-dependent agreement [9]. An accurate forecast can coexist with non-unique physiological parameters, while an identifiable model can still have the wrong biology. Both prediction and parameter interpretation must be assessed.

Clinical validation should subsequently use a prespecified population, prediction horizon and analysis, with external testing before prospective, blinded forecasting. Imaging endpoints should account for treatment effects and pseudoprogression using a defined response-assessment framework such as RANO 2.0 [23]. Agreement with a later MRI or recurrence location is evidence about prediction under the observed treatment; it does not establish the outcome under an alternative treatment. Demonstrating clinical utility would require prospective comparative evaluation of decisions and, ultimately, controlled trials assessing survival, neurological function, quality of life and toxicity.

4.6 Treatment evaluation and candidate strategies

The present simulations suggest why treatment evaluation should consider viable burden and spatial infiltration together: similar threshold-defined volumes can conceal different modeled interiors. Their difference is not a treatment effect, because no therapy was simulated. Extending the framework would require an explicit resection geometry and residual disease state, spatially delivered radiation and drug exposure/response terms. Surgery may also change anatomy and perfusion, requiring updated transport and observation models. Natural growth and treatment-response parameters should be estimated jointly from appropriately timed observations, or constrained independently when they cannot be separated.

An oxygen-dependent radiation-survival model could test the consequences of hypoxia and reoxygenation between fractions, but the relative availability o first needs a calibrated connection to the relevant radiobiology. Neither c nor z currently represents a measured clonogenic population or radiation-induced normal-tissue injury. Computing a tumor-control probability would require additional assumptions about clonogenic fraction and survival; evaluating toxicity would require a separate normal-tissue model. Treatment-related necrosis and imaging changes must be distinguished from spontaneous hypoxic loss. These additions are proposed mechanisms requiring independent experiments, not interpretations already supported by the three simulated fields.

Recent studies illustrate complementary evaluation targets. Balcerak et al. evaluated imaging- and physics-informed infiltration estimates against subsequent recurrence in a dataset of 152 glioblastoma patients [24]. Häger et al. examined model-informed target delineation and voxel-based dose planning in three cases, demonstrating planning feasibility while identifying difficulties in reproducing complex dose distributions [25]. These retrospective prediction and planning studies do not establish a survival benefit from delivering the proposed alternatives. Earlier risk-aware radiotherapy optimization used a cohort of 100 virtual patients [26], illustrating how uncertainty can inform simulated treatment comparisons without constituting a clinical trial.

A next experimental stage could use paired virtual cohorts to compare a specified standard-treatment comparator with adaptive fractionation, spatial dose painting, hypoxia-modifying combinations and alternative drug/radiation sequences. Each candidate would require an explicit

mechanism and independently constrained response parameters. Comparisons should use matched dose/resource constraints, assess normal-brain exposure and delivery feasibility, and test whether a favorable ranking persists across parameter and model uncertainty. Adaptive policies must use only observations available at the decision time. Outcomes could include viable burden, spatial recurrence and time to a prespecified progression criterion; a survival prediction would require a separately validated link to that endpoint. Controlled preclinical oxygen/perfusion perturbations could test the proposed feedback before patient translation. Virtual cohorts should prioritize experimentally testable hypotheses and informative measurements; they cannot identify an effective new therapy solely from a favorable simulated result.

4.7 Limitations and staged roadmap

The anatomical comparison uses one healthy phantom, one seed and one primary parameter set. The ablations and refinement runs are not a population analysis. The recovery experiments vary noise around two fixed states, including one BrainWeb seed; they do not sample between-patient heterogeneity or calibrate the dense-seed oxygen–necrosis comparison. The 3 mm primary BrainWeb fit shares tissue geometry with its generator, and one 2 mm sensitivity example is not a distributional audit of anatomy uncertainty. Scalar migration omits fiber orientation, and no displacement, edema, vascular evolution, immune interactions, phenotype switching or therapy is modeled. Quasi-steady oxygen identifies effective coefficient groups rather than absolute oxygen equilibration rates. Minimum oxygen and necrotic accumulation are under-resolved, and zero clearance accentuates accumulation. Finer simulation grids do not create additional measurement information.

The following sequence makes the roadmap concrete, building on the numerical comparisons and conditional synthetic recovery reported here.

1. **Establish an estimable MRI model.** Resolve the coupled-model numerical limitations, calibrate registered imaging/field observation operators and extend the demonstrated two- and three-rate recovery across seed locations, uncertain initialization and plausible anatomical representations. Add oxygen/death parameters only when independent measurements or justified priors constrain them.
2. **Broaden uncertainty assessment.** Extend the present conditional intervals to propagate initialization, registration, tissue-label and imaging-model uncertainty, explicitly testing the anatomy sensitivity found here. Audit coverage across settings, assess bootstrap calibration and, for Bayesian inference, prior sensitivity, while separating numerical and model errors.
3. **Demonstrate external forecasting.** Fit early scans, freeze the analysis and predict later observations; compare separately calibrated model variants and simple baselines, then repeat on an external cohort with independent spatial biological targets.
4. **Evaluate treatment hypotheses.** Implement and verify treatment and normal-tissue effects, calibrate their response parameters, and compare feasible strategies in virtual cohorts and controlled experiments, including informative imaging schedules.
5. **Test clinical utility.** Prespecify prospective, blinded forecasts before evaluating model-guided decisions in comparative studies and controlled trials with patient-centered benefit and harm endpoints.

Add complexity only to address predictive discrepancies supported by independent observations. Each stage requires its own evidence.

4.8 Conclusion

A minimal oxygen–necrosis extension can be implemented alongside Fisher–KPP within the same anatomical and interactive workflow. Under the assigned conditions, the extension reduced viable burden much more than low-threshold extent, largely through growth inhibition. Its localized

necrotic predictions were sensitive to grid spacing and cannot be interpreted as validated tissue fractions. Staged synthetic experiments estimated homogeneous and tissue-specific baseline rates with conditional uncertainty and withheld forecasts. A proliferation interval miss was retained, and anatomy/grid changes produced a material gray-matter rate shift that noise-based intervals do not capture. Together, the code, recorded experiments and manuscript provide a tested starting point for measurement-based inference and staged validation; patient calibration and independent biological evaluation remain necessary.

Code, data and supplementary material

The manuscript, source code, interactive Jupyter notebooks, recorded simulations, synthetic recovery observations and synchronized visualizations are available at https://github.com/arvidl/brainweb_fisher_kpp. The manuscript project includes LaTeX/BibTeX sources, simulation and figure-generation scripts, machine-readable experiment records, source and data hashes, and numerical-test results. Repository history and the recorded source hashes identify the implementation used for this study. An immutable release identifier and archival DOI should accompany a submitted version when available. BrainWeb attribution and source records are retained separately from the software.

Study scope and declarations

All experiments are computer simulations on anatomical or homogeneous phantoms. No patient measurements, new human sampling or new animal experiments were used. Parameter recovery used synthetic measurements only; patient calibration and external biological validation remain proposed work. Authorship, affiliations, funding, competing interests, contributions and any required AI-use declaration must be finalized before journal submission.

A Anatomical data and initialization

The files are `data/t1_icbm_normal_1mm_pn0_rf0.nii.gz` and `data/phantom_1.0mm_normal_crisp.nii.gz`. The former is the BrainWeb T1 alias with 1 mm spacing, zero added noise and zero intensity nonuniformity; the latter is the matching original normal crisp phantom, converted from MINC to NIfTI. These are not the separate BrainWeb 20-subject anatomical models. Both native arrays have shape $181 \times 217 \times 181$ and the same RAS+ affine

$$A = \begin{pmatrix} 1 & 0 & 0 & -90 \\ 0 & 1 & 0 & -126 \\ 0 & 0 & 1 & -72 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{8}$$

Native gray- and white-matter voxel counts are 902,912 and 674,777. Tissue labels 2, 3 and 8 define the active domain; label 8 uses gray-matter coefficients. CSF is label 1 and is excluded. No tumor is present in the source images. T1 intensities are divided by their 99.8th percentile and clipped for background display only; they do not determine transport or growth coefficients. At 2 mm the affine diagonal is $(2, 2, 2, 1)$ with the same translation, and array shape is $91 \times 109 \times 91$.

The SHA-256 checksums of the bundled compressed files are:

T1: `aa40262b2702653188fd51bdec844a393b825650fe1155a84d8606f5dfd99eb3`
Labels: `afdff78fc1b06ad7c7d8216cf96be737a1450ef5b6abf27eb732c0bf8b66fac0`.

The manifest retains official aliases and source URLs. Regenerating a MINC-to-NIfTI conversion can change compression/header encoding and therefore file hashes; a replacement requires an explicit voxel/affine comparison and updated manifest. BrainWeb attribution and terms remain applicable to these data.

For $\boldsymbol{x}_0 = (-30, 25, 30)$ mm and $\sigma_k = 6$ mm, let $r^2 = \sum_{k=1}^{3}[(x_k - x_{0k})/\sigma_k]^2$. The initialization is

$$c(\boldsymbol{x}, 0) = 80{,}000 \exp(-r^2/2)\, \mathbf{1}_{\{r^2 \le 9\}}\, \mathbf{1}_{\Omega}(\boldsymbol{x}), \qquad z(\boldsymbol{x}, 0) = 0. \tag{9}$$

Oxygen is initialized by the elliptic solve. The physical center is retained when grids change. The nearest saved slice indices on the 2 mm grid are $(30, 76, 51)$, corresponding to RAS $(-30, 26, 30)$ mm: the saved coronal plane is at 26 mm, rather than the off-grid 25 mm seed coordinate. Figure 6 supplies orthogonal views of the same fields as Figure 1.

B Numerical implementation and verification

B.1 Conservative transport and reaction splitting

For neighboring active cells i, j on a Cartesian grid of spacing h, use the harmonic face coefficient

$$D_{ij} = \begin{cases} 2D_iD_j/(D_i + D_j), & D_i + D_j > 0, \\ 0, & \text{otherwise}, \end{cases} \qquad w_{ij} = D_{ij}/h^2. \tag{10}$$

Faces leaving the active domain are assigned zero weight, as are external volume faces. A cell-transport step is

$$c_i^* = c_i^n + \Delta t \sum_{j \in \mathcal{N}(i)} w_{ij}(c_j^n - c_i^n), \qquad \Delta t \le \frac{0.9}{\max_i \sum_j w_{ij}}. \tag{11}$$

Symmetric fluxes conserve total cell count under pure diffusion. The step bound is sufficient for nonnegative explicit updates. Faces are stored without periodic wrapping; no density clipping

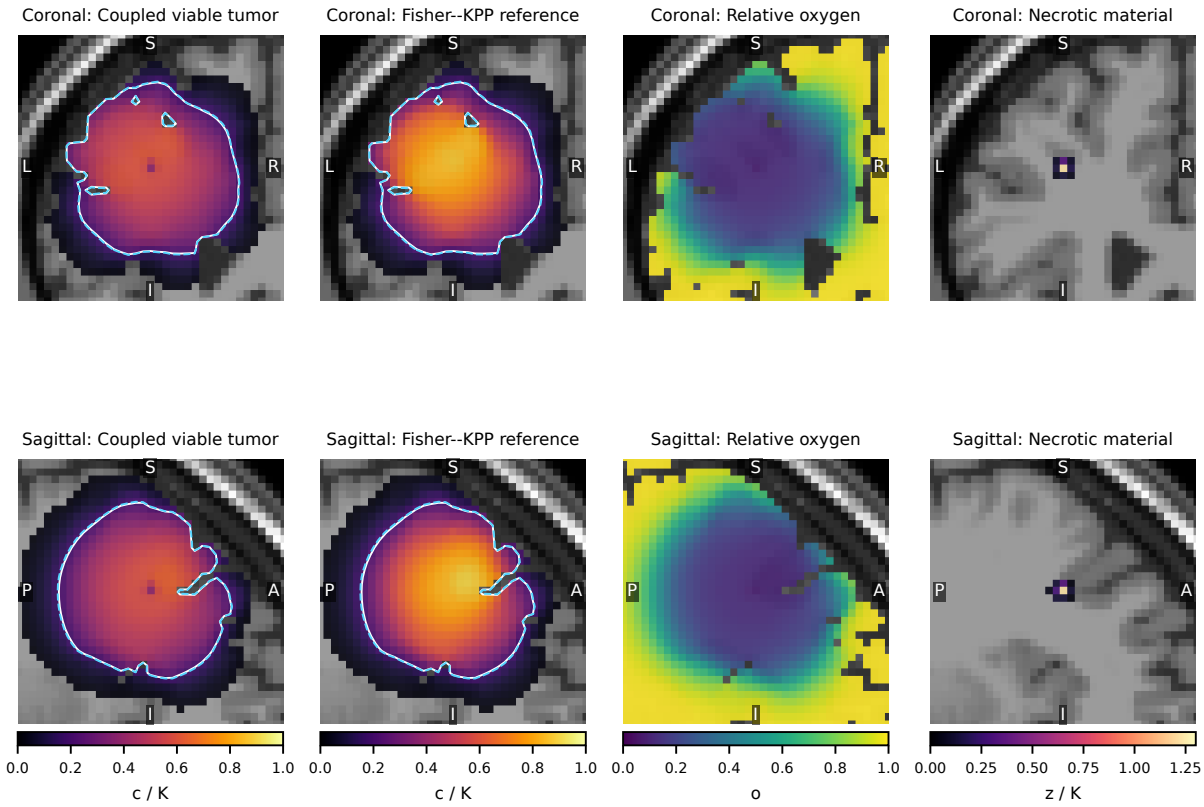


Figure 6: Orthogonal day-90 comparisons through the saved seed planes: coronal $y = 26$ mm (top) and sagittal $x = -30$ mm (bottom). Each panel spans 72×72 mm with native 2 mm pixels. Columns show coupled viable density, the matched Fisher–KPP reference, relative oxygen and necrotic material. The viable and reference scales are identical, and the necrotic scale extends to 1.3. White solid and cyan dashed contours indicate the respective $c/K = 0.2$ boundaries.

is applied. Requested steps are subdivided as needed, and steps end at the exact saved output times.

For M0, the reaction following diffusion is the exact logistic flow

$$c_i^{n+1} = \frac{c_i^*}{\exp(-\rho_i \Delta t) + (c_i^*/K_i)[1 - \exp(-\rho_i \Delta t)]}. \tag{12}$$

For M2, freeze oxygen at o^n and necrosis at z^n during birth. Set $F_i = K_i - z_i^n$. If $F_i > 0$ and $c_i^* < F_i$, use

$$r_i = \rho_i g(o_i^n) F_i / K_i, \qquad c_i^b = \frac{c_i^*}{\exp(-r_i \Delta t) + (c_i^*/F_i)[1 - \exp(-r_i \Delta t)]}. \tag{13}$$

Otherwise $c_i^b = c_i^*$. With $q_i = \delta h(o_i^n)$, the subsequent death/clearance step is

$$L_i = c_i^b[1 - \exp(-q_i \Delta t)], \quad c_i^{n+1} = c_i^b - L_i, \quad z_i^{n+1} = (z_i^n + L_i)\exp(-\lambda_z \Delta t). \tag{14}$$

The same L_i leaves c and enters z, preserving cell equivalents when clearance is zero. Exact frozen-coefficient substeps do not make the full coupled update exact: diffusion, birth, death and clearance are Lie-split, and oxygen is lagged. The complete method is first order in time. The centered spatial stencil has second-order consistency in smooth regions, without a global spatial-order claim at sharp tissue boundaries. The BrainWeb tumor and necrosis arrays use float32 and compiled Numba CPU kernels; both synthetic inference wrappers use float64 for stable finite-difference sensitivities.

B.2 Quasi-steady oxygen solve

Let L_h be the unit-diffusivity no-flux Laplacian on the active voxel graph, $u = c/K$, and $p_i = [1 + \alpha(c_i + z_i)/K_i]^{-1}$. After initialization and each tumor update, solve

$$\left[-\ell_o^2 L_h + \mathrm{diag}(p + \beta u)\right] \boldsymbol{o} = \boldsymbol{p}. \tag{15}$$

The positive reaction diagonal makes this system symmetric positive definite, including disconnected tissue components. Matrix-free conjugate gradients use float64 arithmetic, the previous oxygen state as the initial guess and a Jacobi preconditioner. The relative residual tolerance is 10^{-7}, with a 200-iteration maximum per solve. Nonconvergence or oxygen-bound failure raises an error. The zero-uptake and zero-diffusion limits have direct solutions. Stored oxygen is float32. Outside-domain zeros are excluded-domain sentinels and are excluded from oxygen statistics and overlays.

The primary M2 run required 361 solves and 873 total conjugate-gradient iterations; the largest recorded relative residual was below 10^{-7}. This residual measures algebraic solution accuracy, not accuracy of the continuum approximation on the chosen grid.

B.3 Recorded checks and refinement comparisons

The 50-test suite includes exact logistic growth; reflecting-boundary diffusion modes; heterogeneous-flux conservation; impermeable interfaces and no wrapping; positivity and unsafe-step rejection; parameter checks; exact output times; data affine/tissue checks; uniform and nonuniform oxygen solutions; oxygen barriers; exact death transfer and clearance; occupied-capacity behavior; Fisher–KPP recovery; live-plus-dead balance; coupled temporal refinement against an independent Runge–Kutta calculation; failed-solver reporting; rendering; and replay compatibility. Eight inference tests additionally check the exact reaction limit, transport/imaging mass conservation, unclipped linear imaging, noiseless two-rate recovery

Table 12: Final relative L^2 differences (percent) against the 0.0625-day M2 calculation on the 2 mm grid.

Δt (days)	Viable tumor	Oxygen	Necrotic material
0.25	0.1270	0.0129	4.238
0.125	0.0423	0.0043	1.386

and withheld forecasts, known-noise covariance scaling, uninformative-data rejection, and invalid inputs. Seven tissue-inference tests additionally check three-rate noiseless recovery and withheld prediction, harmonic-face equivalence, cell conservation and CSF exclusion, physical voxel alignment across grids, rejection of a missing tissue class, stability/input checks and observation-region bounds. Test results are stored in `paper/data/pytest.xml`. Notebook smoke checks and media review are separate interface checks.

For field v, final relative L^2 differences are $\|v_{\Delta t} - v_{0.0625}\|_2 / \|v_{0.0625}\|_2$ on the same 2 mm grid (Table 12). The denominator is nonzero in these cases. These are finite-reference differences, not rigorous continuum error bounds. The 1 mm assessment is reported in Table 3; because the tissue labels also change under resampling, it is not used to infer spatial order.

C Synthetic recovery: observation, optimization and resampling

C.1 Imaging operator and fixed design

The homogeneous box is $[-30, 30]^3$ mm with reflecting boundaries. For known u_0, the initial cell averages are computed from the product of one-dimensional Gaussian integrals using the error function. The fine generator has 60^3 cells and inversion has 30^3 cells. Both map to 10^3 observation voxels per scan. For a Gaussian point-spread kernel G_{σ_b} with $\sigma_b = 2$ mm and an observation voxel V_v of volume 6^3 mm^3,

$$H_v[u(t)] = \frac{1}{|V_v|} \int_{V_v} (G_{\sigma_b} * u)(\boldsymbol{x}, t)\, \mathrm{d}\boldsymbol{x}, \qquad y_{jv} = H_v[u(t_j)] + \epsilon_{jv}, \quad \epsilon_{jv} \sim \mathcal{N}(0, 0.03^2). \tag{16}$$

Implementation uses a discrete Gaussian filter with reflecting extension, followed by aligned block averages. The same signal calibration and blur are assumed known in both models. Noise is independent across observation voxels and scan times and is added after averaging; blurring independently generated noise instead would require a correlated likelihood. Values outside $[0, 1]$ are retained. All training voxels are used without a data-dependent region selection. The cellularity proxy has a known normalized scale and does not identify an absolute K from imaging.

C.2 Maximum likelihood and local information

With $\theta = (\log D, \log \rho)$, using numerical values in mm^2/day and day^{-1}, define

$$r_{jv}(\theta) = \frac{H_v[u(e^{\theta_1}, e^{\theta_2}, t_j)] - y_{jv}}{0.03}, \qquad \widehat{\theta} = \arg\min_{\theta} \tfrac{1}{2} \|\boldsymbol{r}(\theta)\|_2^2. \tag{17}$$

The bounds are $0.05 \le D \le 0.8$ mm^2/day and $0.01 \le \rho \le 0.09$ day^{-1}; they are optimizer constraints, not a prior distribution. Starting pairs are $(0.12, 0.02)$, $(0.30, 0.04)$ and $(0.65, 0.075)$. Bounded trust-region least squares uses a three-point finite-difference Jacobian with relative step 10^{-4} and Jacobian column scaling [16]. Termination tolerances are 10^{-8} for objective/step changes and 10^{-6} for the gradient. The selected fit minimizes training residual sum of squares. Inversion has a fixed maximum time step that is stable throughout the bounds; it does not change with the fitted rate except for landing at exact output times.

Let $J = \partial \boldsymbol{r}/\partial\theta$ at the fit. The singular-value decomposition gives

$$J = USV^{\mathsf{T}}, \qquad C_\theta = VS^{-2}V^{\mathsf{T}}, \qquad I_i = \exp\left[\widehat{\theta}_i \pm 1.95996\sqrt{(C_\theta)_{ii}}\right]. \tag{18}$$

Because measurement variance is known, C_θ is not rescaled by fitted residual variance. The smallest singular value must exceed 10^{-8} times the largest. A fit within 10^{-4} log units of a bound is flagged and its Wald intervals must not be interpreted as calibrated. In the primary experiment, weighted singular values were approximately 107.9 and 18.0. With only $t = 0$, both rate sensitivities vanish, and the fit is rejected as locally unidentifiable.

C.3 Bootstrap, forecasts and repeated-noise audit

For each of 100 bootstrap trials, training observations were generated from the fitted coarse-grid mean plus new Gaussian noise and refitted from the primary estimate [17]. Marginal percentile intervals use the 2.5th and 97.5th percentiles. Every fitted parameter pair was propagated to the five scan times. Pointwise mean-signal bands use trajectory percentiles. Approximate new-scan prediction bands resample these bootstrap trajectories 1,000 times and add independent observation noise before taking percentiles. These constructions are conditional frequentist approximations, not samples from a Bayesian posterior, and do not provide simultaneous spatial coverage.

The separate coverage audit generated 60 noisy training datasets from the fine-grid signal at the known parameters. Each fit started from $(0.25, 0.04)$. Bias is the mean estimate minus the known value; RMSE is the square root of mean squared estimation error. Coverage counts intervals containing the known rate. Exact two-sided binomial 95% intervals quantify sampling uncertainty in these fractions. The audit concerns local Jacobian intervals; bootstrap coverage would require a separate repeated or nested study. All optimization failures or boundary fits stop execution rather than being discarded. There were none in the recorded experiment.

NumPy seed 20261017 spawns four independent streams for the primary noise, bootstrap, repeated-noise audit and predictive noise. The known truth is used to evaluate recovery and forecasts, not to select fitted parameters or tune uncertainty. Unknown age, seed geometry, noise variance, capacity, registration, observation calibration and model discrepancy are not inferred. The mesh/time comparison uses the same PDE family and is a numerical sensitivity check, not validation against a different biological mechanism.

D BrainWeb recovery: tissue rates and anatomy sensitivity

D.1 Domain, seed and physical observation operator

The full 3 mm BrainWeb volume has the affine translation of Appendix A and diagonal spacing $(3, 3, 3)$ mm. The active set contains labels 2, 3 and 8; its 58,669 voxels evolve using the conservative harmonic-face kernel with float64 density. Outside-domain values remain zero, and no cells cross CSF or extracerebral boundaries. The normalized model is

$$\partial_t u = \nabla\cdot\left(D(\boldsymbol{x})\nabla u\right) + \rho u(1-u), \qquad D(\boldsymbol{x}) = \begin{cases} D_{\mathrm{WM}}, & \text{label 3,} \\ D_{\mathrm{GM}}, & \text{labels 2 or 8,} \end{cases} \tag{19}$$

with $u_0(\boldsymbol{x}) = 0.2\exp[-\frac{1}{2}\sum_k((x_k - x_{0k})/6)^2]\mathbf{1}_\Omega$, where $\boldsymbol{x}_0 = (-30, 25, 30)$ mm. The analytic Gaussian is point-sampled and untruncated; it differs from both the integrated homogeneous initialization and the dense, truncated anatomical forward seed. Its absolute cell scale is known only by assumption; K is not estimated.

Observation centers along each axis are

$$x_{v,k} = a_k + (v + \tfrac{1}{2} - 5)6 \text{ mm}, \qquad v = 0, \dots, 9, \qquad \boldsymbol{a} = (-30, 24, 30) \text{ mm}. \tag{20}$$

Let $\mathcal{I}_h$ be trilinear interpolation in RAS coordinates and $G_{2,h}$ the discrete 2 mm Gaussian filter. Approximate voxel averaging uses the fixed midpoint offsets $s_l \in \{-2, 0, 2\}$ mm:

$$H_v^h[u] = \frac{1}{27} \sum_{l_1,l_2,l_3=1}^{3} \mathcal{I}_h[G_{2,h}u](\boldsymbol{x}_v + (s_{l_1}, s_{l_2}, s_{l_3})), \qquad y_{jv} = H_v^h[u(t_j)] + \epsilon_{jv}, \quad \epsilon_{jv} \sim \mathcal{N}(0, 0.02^2). \tag{21}$$

The filter uses zero extension outside the image, with the tumor field zero outside parenchyma. Noise is independent after imaging and is never clipped. Physical locations and quadrature remain fixed across 3 mm and 2 mm grids. This is a calibrated normalized-cellularity proxy; T1 intensity is not the tumor observation. No fitted or true-signal threshold selects the FOV or near-seed evaluation region.

D.2 Three-rate likelihood, uncertainty and controls

The log-rate vector is $\theta = (\log D_{\mathrm{WM}}, \log D_{\mathrm{GM}}, \log \rho)$, with numerical rates in the stated units. Minimize the squared residuals $(H_v^h[u(t_j)] - y_{jv})/0.02$ at days 20, 40 and 60. Search bounds are $(0.05, 0.02, 0.01)$ to $(1.0, 0.6, 0.09)$, and starts are $(0.25, 0.20, 0.025)$, $(0.45, 0.12, 0.04)$ and $(0.80, 0.05, 0.065)$. No tissue-ratio equality, ordering or probability prior is imposed. The finite-difference settings, log-covariance construction, singular-value rank criterion and boundary flag follow Appendix C, now with three columns. A sufficient diffusion bound is $6 \max(D_{\mathrm{WM}}, D_{\mathrm{GM}})\Delta t/h^2 \le 0.9$, satisfied throughout the search region. Steps land exactly at scan times.

The primary generator uses the same 3 mm geometry with a 0.125-day maximum step; inversion uses 0.5 day. Bootstrap training samples are drawn from the fitted inverse mean and refitted from its estimate. The 60 independent coverage trials instead use the fine-time-step known-rate mean, each starting from $(0.4, 0.15, 0.035)$. No failed or boundary trial is omitted; either stops the notebook. Time-zero data are rejected because all three sensitivities vanish. Missing-tissue rank deficiency is also tested in the software suite.

NumPy seed 20261018 spawns independent primary, bootstrap, coverage and predictive-noise streams. Each of 100 fitted bootstrap triples is propagated to all five scan times; resampling these trajectories 1,000 times and adding new independent noise gives approximate pointwise new-scan prediction intervals. The withheld days do not enter any fit. Whole-FOV metrics use 1,000 voxels, while near-seed metrics use the 120 voxel centers within a prespecified 18 mm radius of $\boldsymbol{x}_0$. These spatial fractions should not be read as independent coverage audits.

D.3 Separate anatomy/grid perturbation

The sensitivity generator uses the full $91 \times 109 \times 91$ BrainWeb volume at 2 mm and a 0.125-day step, keeping physical rates, analytic seed definition, imaging settings and observation locations fixed. The 3 mm inverse model fits its noise-free mean and a paired noisy dataset with exactly the primary noise draw added. The noiseless residual objective uses the same constant noise weight, which only rescales the objective; no sampling interval is interpreted for that deterministic fit. Both fits begin at $(0.4, 0.15, 0.035)$.

Nearest-neighbor tissue resampling changes the interface graph. Point sampling of the Gaussian and the discrete blur/interpolation also change, so the fitted shifts combine anatomy and numerical representation. The test is neither a pure mesh-convergence experiment nor an empirical distribution of segmentation errors. Halving the inverse time step on the fixed 3 mm

anatomy is reported separately. No patient, oxygen or necrosis parameter is inferred in these experiments.

E Inverse formulation, priors and identifiability

E.1 An example estimation objective

With the observation model in Equation (7), a possible maximum-a-posteriori objective for continuous approximately Gaussian observations is

$$(\widehat{\boldsymbol{\theta}}, \widehat{\boldsymbol{\eta}}) = \arg\min_{\boldsymbol{\theta},\boldsymbol{\eta}} \ \mathcal{J}(\boldsymbol{\theta}, \boldsymbol{\eta}), \tag{22}$$

$$\mathcal{J} = \frac{1}{2}\sum_{m,j} \boldsymbol{r}_{mj}^{\mathsf{T}} \Sigma_{mj}^{-1} \boldsymbol{r}_{mj} - \log \pi(\boldsymbol{\theta}, \boldsymbol{\eta}), \qquad \boldsymbol{r}_{mj} = \mathcal{H}_m[\boldsymbol{s}_{\boldsymbol{\theta}}(t_j); \boldsymbol{\eta}_m] - \boldsymbol{y}_{mj}, \tag{23}$$

where $\boldsymbol{s} = (c, o, z)$ and Σ models measurement covariance. Cross-time or cross-modality correlations require a joint likelihood rather than independent terms. Segmentations may need a probabilistic classification likelihood rather than Gaussian residuals. Registration, partial volume, scanner bias, initial fields and uncertain state-to-signal mappings should enter the likelihood or be propagated as nuisance uncertainty. Bayesian sampling can quantify uncertainty; an optimizer alone does not provide calibrated predictive intervals. No coupled-model or patient-data maximum-a-posteriori fit was performed; the synthetic analyses in Appendices C and D used maximum likelihood without a prior.

E.2 Important structural and practical ambiguities

1. **Absolute cell scale.** In terms of $u = c/K$ and $w = z/K$ with uniform K, the normalized dynamics do not determine K. Scaling (c, z, K) together leaves (u, w, o) unchanged. An observation dependent only on normalized density cannot identify the absolute cell count without external calibration.
2. **Age and rates.** For an unknown elapsed tumor age, multiplying D, ρ, δ and λ_z by a common factor and dividing the elapsed time by that factor leaves the autonomous state trajectory at the observation unchanged. Known longitudinal time intervals can break this particular scaling, although seed and parameter confounding may remain.
3. **Oxygen-growth coupling.** Where all observed cells remain below θ_p, the birth term depends on ρ/θ_p rather than on those parameters separately, for a fixed oxygen field. Oxygen threshold calibration and observations spanning informative regimes are needed.
4. **Delivery and uptake.** Neglecting spatial oxygen transport locally gives

$$o = \frac{1}{1 + \beta u[1 + \alpha(u + w)]}. \tag{24}$$

   One local state constrains an effective supply–uptake combination, not separate mechanisms. Spatially varying, calibrated oxygen observations across burdens could help, but do not guarantee recovery. An exchange length far below measurement and mesh resolution can be practically unidentifiable.
5. **Quasi-steady reduction.** If a dimensional oxygen equation is divided by a baseline restoration rate, ℓ_o^2 and β represent coefficient ratios. Steady observations alone cannot identify an independent absolute oxygen relaxation time removed by that reduction.
6. **Necrotic history.** For known $q = \delta h(o)$,

$$z(t) = z(0)e^{-\lambda_z t} + \int_0^t q(s)c(s)e^{-\lambda_z(t-s)}\,\mathrm{d}s. \tag{25}$$

A single terminal necrotic measurement cannot generally separate formation history and clearance. Zero clearance is a model choice here, not an estimated absence of biological removal.

Positive parameters can be transformed as $\log \theta_k \sim \mathcal{N}(\mu_k, \sigma_k^2)$, with dimensional reference units stated. For thresholds, use $\theta_p = \text{logit}^{-1}(a)$ and $\theta_d = \theta_p \, \text{logit}^{-1}(b)$. A zero-clearance component or other exact-zero ablation requires an explicit point mass or a separate model, rather than a continuous positive prior. Neither prior hyperparameters nor population distributions are supplied by the current phantom experiment.

Before a full fit, evaluate scaled observation sensitivities, their singular values, profile likelihoods and posterior dependence across plausible initial states and observation mappings. The number of voxels is not the number of independent constraints. Begin with a global migration scale and growth coefficient, restrict other quantities using external information, and add freedom only when independent observations support it. Identifiability and predictive comparison should be repeated for each candidate model and intended measurement set.

F Reproducibility and interactive artifacts

`paper/run_experiments.py` runs the nine assigned-parameter cases, verifies the bundled data and checks the primary trajectory against the saved example. Under `paper/data/`, it records metrics/settings, time courses, lossless seed-plane fields and provenance. Source hashes cover the model, data loader, renderer, configuration, lockfile and experiment script. Because simulation preceded its source commit, these working-file hashes identify the implementation independently of the earlier base revision.

`paper/make_figures.py` reads these records to generate figures, table rows and numerical macros used directly by the manuscript. The local Python environment is defined by `pyproject.toml` and `uv.lock`. A `Makefile` rebuilds the figures and PDF, with separate targets to rerun the anatomical experiments and synthetic recovery. The native and 2 mm anatomical examples are deterministic; they do not sample biological uncertainty.

`synthetic_parameter_recovery.ipynb` contains self-contained homogeneous (Part A) and BrainWeb (Part B) experiments using `src/brainweb_kpp/inference.py` and `src/brainweb_kpp/tissue_inference.py`, respectively, with the shared transport/reaction kernel. `scripts/run_recovery_notebook.py` runs and saves both. Under `outputs/synthetic_recovery/` and `outputs/brainweb_recovery/`, each `summary.json` records settings, units, seeds, intervals, errors, coverage, software versions and source/data hashes. Each `recovery_data.npz` retains observations, parameter samples, interval arrays and forecast bands. Part B additionally saves 2 mm stress observations and numerical-reference predictions. Five and six PNG figures, respectively, accompany these arrays. Scoped code-cell hashes distinguish the two experiments independently of notebook output metadata. `paper/make_recovery_figures.py` and `paper/make_brainweb_recovery_figures.py` check source, code and array hashes and regenerate vector figures, numerical macros and tables. The BrainWeb generator also recomputes coverage, bias, forecast metrics and numerical discrepancies from the saved arrays. The full reproduction target reruns the nine assigned-parameter forward cases and both recovery experiments before compiling the PDF.

`tumor_growth_explorer.ipynb` exposes both models; `coupled_example.ipynb` presets the 90-day comparison. Time and RAS slices are shared across the four-field view. Live histories permit arbitrary 3D slicing, while compact replay files retain the three seed-plane histories and final full volumes. The slower comparison MP4 uses 400 frames at 20 frames/s, including a 120-frame day-90 hold; playback duration is distinct from simulated time. Synchronized movies help inspect output but are not quantitative validation. Manuscript figures retain the full computed necrotic range, unlike a display limited to 0–1.

References

[1] Ronald A. Fisher. The wave of advance of advantageous genes. *Annals of Eugenics*, 7(4): 355–369, 1937. doi:10.1111/j.1469-1809.1937.tb02153.x. URL https://doi.org/10.1111/j.1469-1809.1937.tb02153.x.

[2] Andrey N. Kolmogorov, Ivan G. Petrovskii, and Nikolai S. Piskunov. étude de l'équation de la diffusion avec croissance de la quantité de la matière et son application à un problème biologique. *Bulletin de l'Université d'État à Moscou, Série Internationale A*, 1:1–25, 1937. URL https://math.i-learn.unito.it/mod/resource/view.php?forceview=1&id=73373. Original French article; link provides an English translation.

[3] Kristin R. Swanson, Ellsworth C. Alvord, Jr., and James D. Murray. A quantitative model for differential motility of gliomas in grey and white matter. *Cell Proliferation*, 33(5): 317–329, 2000. doi:10.1046/j.1365-2184.2000.00177.x. URL https://doi.org/10.1046/j.1365-2184.2000.00177.x.

[4] Andrea Comba, Syed M. Faisal, Maria Luisa Varela, Todd Hollon, Wajd N. Al-Holou, Yoshie Umemura, Felipe J. Nunez, Sebastien Motsch, Maria G. Castro, and Pedro R. Lowenstein. Uncovering spatiotemporal heterogeneity of high-grade gliomas: From disease biology to therapeutic implications. *Frontiers in Oncology*, 11:703764, 2021. doi:10.3389/fonc.2021.703764. URL https://doi.org/10.3389/fonc.2021.703764.

[5] Alicia Martínez-González, Gabriel F. Calvo, Luis A. Pérez Romasanta, and Víctor M. Pérez-García. Hypoxic cell waves around necrotic cores in glioblastoma: A biomathematical model and its therapeutic implications. *Bulletin of Mathematical Biology*, 74(12):2875–2896, 2012. doi:10.1007/s11538-012-9786-1. URL https://doi.org/10.1007/s11538-012-9786-1.

[6] Khushi C. Hiremath, Kenan Atakishi, Ernesto A. B. F. Lima, Maguy Farhat, Bikash Panthi, Holly Langshaw, Mihir D. Shanker, Wasif Talpur, Sara Thrower, Jodi Goldman, Caroline Chung, Thomas E. Yankeelov, and David A. Hormuth, II. Identifiability and model selection frameworks for models of high-grade glioma response to chemoradiation. *Philosophical Transactions of the Royal Society A*, 383(2293):20240212, 2025. doi:10.1098/rsta.2024.0212. URL https://doi.org/10.1098/rsta.2024.0212.

[7] Hugo J. M. Miniere, David A. Hormuth, II, Ernesto A. B. F. Lima, Maguy Farhat, Bikash Panthi, Holly Langshaw, Mihir D. Shanker, Wasif Talpur, Sara Thrower, Jodi Goldman, Sophia Ty, Caroline Chung, and Thomas E. Yankeelov. A data assimilation framework for predicting the spatiotemporal response of high-grade gliomas to chemoradiation. *BMC Cancer*, 25:1239, 2025. doi:10.1186/s12885-025-14557-3. URL https://doi.org/10.1186/s12885-025-14557-3.

[8] Ray Zirui Zhang, Ivan Ezhov, Michal Balcerak, Andy Zhu, Benedikt Wiestler, Bjoern Menze, and John S. Lowengrub. Personalized predictions of glioblastoma infiltration: Mathematical models, physics-informed neural networks and multimodal scans. *Medical Image Analysis*, 101:103423, 2025. doi:10.1016/j.media.2024.103423. URL https://doi.org/10.1016/j.media.2024.103423.

[9] Gianluca Nocera, Francesco Sanvito, Jingwen Yao, Sonoko Oshima, Samuel A. Bobholz, Ashley Teraishi, Catalina Raymond, Kunal Patel, Richard G. Everson, Linda M. Liau, Jennifer Connelly, Antonella Castellano, Pietro Mortini, Noriko Salamon, Timothy F. Cloughesy, Peter S. LaViolette, and Benjamin M. Ellingson. Independent histological validation of MR-derived radio-pathomic maps of tumor cell density using image-guided biopsies in human brain tumors. *Journal of Neuro-Oncology*, 175(1):111–122, 2025. doi:10.1007/s11060-025-05105-x. URL https://doi.org/10.1007/s11060-025-05105-x.

[10] Ayesha Das, David A. Hormuth, II, John Virostko, Patrik Parker, C. Chad Quarles, and Thomas E. Yankeelov. Repeatability of an MRI protocol for generating habitats based on cellularity, perfusion, and hypoxia in a murine model of glioma. *Magnetic Resonance Imaging*, 133:110744, 2026. doi:10.1016/j.mri.2026.110744. URL https://doi.org/10.1016/j.mri.2026.110744. Published online 6 July 2026.

[11] Meryem Abbad Andaloussi, Stephane Urcun, David A. Hormuth, II, Guillermo Lorenzo, Giuseppe Sciume, Cheguye Wu, Thomas E. Yankeelov, and Stephane P. A. Bordas. An MRI-informed poromechanical model for organ-scale prediction of glioma growth. arXiv:2605.05013, 2026. URL https://arxiv.org/abs/2605.05013. Preprint, version 1, 6 May 2026; not peer reviewed in the cited record.

[12] Abdullah Al Noman, Pratyush Kumar Singh, David A. Hormuth, II, and Danial Faghihi. Dynamic image-informed selection of biomechanical tumor growth models. arXiv:2607.04551, 2026. URL https://arxiv.org/abs/2607.04551. Preprint, version 1, 5 July 2026; not peer reviewed in the cited record.

[13] Veronika Hofmann, Pirmin Schlicke, Jan Zawallich, Heiko Enderling, and Christina Kuttler. 2D reaction-diffusion model-based biopsy simulation for dynamic tumor growth parameter estimation. arXiv:2609.22363, 2026. URL https://arxiv.org/abs/2609.22363. Preprint, version 1, 17 September 2026; theoretical validation using simulated biopsies.

[14] D. Louis Collins, Alex P. Zijdenbos, Vasken Kollokian, John G. Sled, Noor J. Kabani, Colin J. Holmes, and Alan C. Evans. Design and construction of a realistic digital brain phantom. *IEEE Transactions on Medical Imaging*, 17(3):463–468, 1998. doi:10.1109/42.712135. URL https://doi.org/10.1109/42.712135.

[15] McConnell Brain Imaging Centre. BrainWeb: Simulated brain database and anatomical model of normal brain. Montreal Neurological Institute, McGill University, n.d. URL https://brainweb.bic.mni.mcgill.ca/anatomic_normal.html. Accessed 17 October 2026.

[16] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17: 261–272, 2020. doi:10.1038/s41592-019-0686-2. URL https://doi.org/10.1038/s41592-019-0686-2.

[17] Bradley Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979. doi:10.1214/aos/1176344552. URL https://doi.org/10.1214/aos/1176344552.

[18] Oliver Eidel, Jan-Oliver Neumann, Sina Burth, Pascal J. Kieslich, Christine Jungk, Felix Sahm, Philipp Kickingereder, Karl Kiening, Andreas Unterberg, Wolfgang Wick, Heinz-Peter Schlemmer, Martin Bendszus, and Alexander Radbruch. Automatic analysis of cellularity in glioblastoma and correlation with ADC using trajectory analysis and automatic nuclei counting. *PLOS ONE*, 11(7):e0160250, 2016. doi:10.1371/journal.pone.0160250. URL https://doi.org/10.1371/journal.pone.0160250.

[19] Corentin Martens, Laetitia Lebrun, Christine Decaestecker, Thomas Vandamme, Yves-Rémi Van Eycke, Antonin Rovai, Thierry Metens, Olivier Debeir, Serge Goldman, Isabelle Salmon, and Gaetan Van Simaeys. Initial condition assessment for reaction-diffusion glioma growth models: A translational MRI-histology (in)validation study. *Tomography*, 7(4):650–674, 2021. doi:10.3390/tomography7040055. URL https://doi.org/10.3390/tomography7040055.

[20] Fatemeh Arzanforoosh, Maaike Van der Velden, Avery J. L. Berman, Sebastian R. Van der Voort, Erwin M. Bos, Joost W. Schouten, Arnaud J. P. E. Vincent, Johan M. Kros, Marion Smits, and Esther A. H. Warnert. MRI-based assessment of brain tumor hypoxia: Correlation with histology. *Cancers*, 16(1):138, 2024. doi:10.3390/cancers16010138. URL https://doi.org/10.3390/cancers16010138.

[21] Graham Pash, Umberto Villa, David A. Hormuth, II, Thomas E. Yankeelov, and Karen Willcox. Predictive digital twins with quantified uncertainty for patient-specific decision making in oncology. *Journal of Computational Physics*, 560:114937, 2026. doi:10.1016/j.jcp.2026.114937. URL https://doi.org/10.1016/j.jcp.2026.114937.

[22] David A. Hormuth, II, Jared A. Weis, Stephanie L. Barnes, Michael I. Miga, Erin C. Rericha, Vito Quaranta, and Thomas E. Yankeelov. Predicting in vivo glioma growth with the reaction diffusion equation constrained by quantitative magnetic resonance imaging data. *Physical Biology*, 12(4):046006, 2015. doi:10.1088/1478-3975/12/4/046006. URL https://doi.org/10.1088/1478-3975/12/4/046006.

[23] Patrick Y. Wen, Martin van den Bent, Gilbert Youssef, Timothy F. Cloughesy, Benjamin M. Ellingson, Michael Weller, Evanthia Galanis, Daniel P. Barboriak, John de Groot, Mark R. Gilbert, Raymond Huang, Andrew B. Lassman, Minesh Mehta, Annette M. Molinaro, Matthias Preusser, Rifaquat Rahman, Lalitha K. Shankar, Roger Stupp, Javier E. Villanueva-Meyer, Wolfgang Wick, David R. Macdonald, David A. Reardon, Michael A. Vogelbaum, and Susan M. Chang. RANO 2.0: Update to the response assessment in neuro-oncology criteria for high- and low-grade gliomas in adults. *Journal of Clinical Oncology*, 41(33):5187–5199, 2023. doi:10.1200/JCO.23.01059. URL https://doi.org/10.1200/JCO.23.01059.

[24] Michal Balcerak, Jonas Weidner, Petr Karnakov, Ivan Ezhov, Sergey Litvinov, Petros Koumoutsakos, Tamaz Amiranashvili, Ray Zirui Zhang, John S. Lowengrub, Igor Yakushev, Benedikt Wiestler, and Bjoern Menze. Individualizing glioma radiotherapy planning by optimization of a data and physics-informed discrete loss. *Nature Communications*, 16(1): 5982, 2025. doi:10.1038/s41467-025-60366-4. URL https://doi.org/10.1038/s41467-025-60366-4.

[25] Wille Häger, Iuliana Toma-Daşu, Mehdi Astaraki, and Marta Lazzeroni. Treating the invisible target: treatment planning targeting the microscopic tumor spread in gliomas. *Frontiers in Oncology*, 16:1872089, 2026. doi:10.3389/fonc.2026.1872089. URL https://doi.org/10.3389/fonc.2026.1872089.

[26] Anirban Chaudhuri, Graham Pash, David A. Hormuth, II, Guillermo Lorenzo, Michael Kapteyn, Chengyue Wu, Ernesto A. B. F. Lima, Thomas E. Yankeelov, and Karen Willcox. Predictive digital twin for optimizing patient-specific radiotherapy regimens under uncertainty in high-grade gliomas. *Frontiers in Artificial Intelligence*, 6:1222612, 2023. doi:10.3389/frai.2023.1222612. URL https://doi.org/10.3389/frai.2023.1222612.